Can an MLP Absorb Its Skip Connection Exactly?

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Abstract

Residual networks add each block's input to its output; skipless ones now train as well. Can a block without biases absorb its skip: can a skipless block of the same kind compute the same function? Here we show that almost every ReLU or GELU block of at least d_{model} units absorbs its skip with exactly d_{model} more units, with d_{model} the model dimension, while no number of squared-ReLU, gated (ReGLU, SwiGLU, GeGLU) or $\tanh$ units or attention heads does. Without normalization, no skipless squared-ReLU, gated or $\tanh$ network equals a residual one, and a skipless ReLU or GELU network needs at least $2d_{\text{model}}$ units in its first block to equal almost any residual one; yet every skipless network is a limit of residual ones of equal widths. Under RMSNorm, almost no residual network equals any skipless one. These results say when a skip can be traded for hidden units.

1 Introduction

Almost every deep network trained today, from residual networks (He et al., 2016) to transformers (Vaswani et al., 2017), adds the input of each block to its output through a skip connection. The usual reason is training: the sum gives the gradient a short path through the network, and it makes the identity map easy to express (Hardt and Ma, 2017; Veit et al., 2016).

Skipless networks now train too. He and Hofmann (2024) remove the skips of transformer blocks without losing training speed, and Ji et al. (2026) train a skipless ViT-B on ImageNet that outperforms the residual one. Can a skip let a block compute a function that no skipless block of the same kind computes?

We answer this question block by block. Take an MLP block B with m hidden units, or an attention block with H heads, on hidden states of dimension d_{model}; its width is m or H. Two blocks are of the same kind if both are vanilla MLP blocks with the same activation, both are gated with the same gate, or both are attention blocks. Is the residual block $x \mapsto x + B(x)$ also a skipless block of the same kind, of the same width or wider? If it is, we say that B *absorbs* its skip. The skip is worth n units if B absorbs it with n extra units but not with $n-1$; if no width suffices, it is worth more than any number of units.

Without biases, the answer depends on the activation.

- For almost every ReLU or GELU block with at least d_{model} units, a skip is worth exactly d_{model} units: d_{model} new units, each mirroring an existing unit, absorb it, and fewer do not (Theorem 3.1(ii)).
- For ReLU^2 blocks, whose activation is the square of ReLU, gated blocks (ReGLU, SwiGLU, GeGLU) and $\tanh$ blocks, no number of units absorbs a skip (Theorem 3.3).
- For attention blocks, no number of heads absorbs a skip (Theorem 3.4).

Author contributions. Marko Karbevski (MK) conceived and led the study. Antonij Mijoski (AM) proved the absorption result for a single ReLU block (Theorem 3.1(i), Theorem A.5), the difficult case, paving the way forward. MK extended the tools to GELU (Theorem A.7), proved the cases of ReLU^2 and of the gated variants ReGLU, SwiGLU and GeGLU (Theorems 3.3 and 4.1) and the results at depth (Theorem 4.2 and Proposition 4.3). Both authors worked on the writing. All of MK's work on the results and the proofs was done with AI assistants.

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Outside Appendix C, blocks have no biases, as in many current language models (Team OLMo et al., 2025; Yang et al., 2025). With biases, every ReLU^2 and gated block absorbs its skip with more units, while $\tanh$ and attention blocks still never do (Appendix C).

Four one-line reasons. Follow the input along a ray, $x = tu$ with $t > 0$. The skip adds tu, which grows like t. A ReLU^2 block grows like t^2. A gated block, whose gate vanishes at 0, is of order t^2 near the origin. A $\tanh$ block stays bounded. So none of them can produce the term tu (Figure 1, Section 3.2). Attention mixes the tokens of a sequence, while a skip leaves each token where it is.

Networks. Throughout, a network starts with a linear embedding and ends with a linear readout. In a pre-RMSNorm network, every block and the readout read their input through RMSNorm, which divides each token by its root mean square, regularized by a constant $\varepsilon \geq 0$. Without normalization, no residual network of ReLU^2, gated or $\tanh$ blocks equals a skipless network of any widths (Theorem 4.1), and a skipless ReLU or GELU network needs at least $2d_{\text{model}}$ units in its first block, whatever its later widths, to equal almost any residual one (Theorem 4.2). Under RMSNorm, almost no residual network, of any blocks, equals a skipless network of any widths (Theorem 4.5): as an input is scaled by $t \to \infty$, the output of a skipless network converges at rate $1/t^2$ or faster, and that of almost every residual network only at rate $1/t$.

No function lost in the limit. Without normalization, every skipless network is a limit of residual networks of the same widths, uniformly on bounded sets of inputs, as their weights diverge (Proposition 5.1(i)): the skip adds functions and, in the limit, removes none. Table 1 collects the results; Section 2 fixes the setting, Section 3 treats one block, Section 4 networks and Section 5 limits. Proofs are in the appendix.

2 Setup

One equation states the question: can a block absorb a linear term?

Blocks. A hidden state is a matrix $h \in \mathbb{R}^{T \times d_{\text{model}}}$ whose rows are the T tokens of a sequence. An MLP block acts on each token separately. On a row $x \in \mathbb{R}^{1 \times d_{\text{model}}}$ of a hidden state it computes

$$B_\theta(x) = f(xW_{\text{read}})\, W_{\text{write}}, \qquad \theta = (W_{\text{read}}, W_{\text{write}}), \tag{1}$$

with a read W_{read} and a write W_{write}. In a vanilla block with m hidden units, $W_{\text{read}} \in \mathbb{R}^{d_{\text{model}} \times m}$, $W_{\text{write}} \in \mathbb{R}^{m \times d_{\text{model}}}$, and f applies an activation σ to each unit: ReLU, GELU, $\tanh$, or the square of ReLU, ReLU^2. In a gated block the read has two halves, $W_{\text{read}} = [\, W_{\text{g}} \mid W_{\text{v}} \,]$, and unit i multiplies its gate $\sigma\big((xW_{\text{g}})_i\big)$ by its value $(xW_{\text{v}})_i$, with σ the ReLU in ReGLU, SiLU in SwiGLU and GELU in GeGLU (Shazeer, 2020). Attention blocks follow in Section 3.3.

The question as an equation. A skip adds x to the output of a block. More generally, can a block absorb a linear term xM, for a $d_{\text{model}} \times d_{\text{model}}$ matrix $M \neq 0$? We look for parameters θ' of a block of the same kind, and of any width, such that

$$xM + B_\theta(x) = B_{\theta'}(x) \qquad \text{for all } x \in \mathbb{R}^{1 \times d_{\text{model}}}. \tag{2}$$

The skip is $M = I$. Other matrices cover learned linear skips and, below, networks.

Networks. A network maps an input $X \in \mathbb{R}^{T \times d_{\text{in}}}$ to

$$\mathcal{M}(X) = C \circ D \circ (A_L + B_L) \circ \cdots \circ (A_1 + B_1) \circ E\,(X),$$

where E is the linear embedding $X \mapsto XE$, with $E \in \mathbb{R}^{d_{\text{in}} \times d_{\text{model}}}$, $B_1, \ldots, B_L$ are $L \geq 1$ blocks, D is the linear readout, or unembedding, $h \mapsto hD$, with $D \in \mathbb{R}^{d_{\text{model}} \times d_{\text{out}}}$, A_ℓ is the skip of block ℓ, the zero map in a *skipless* network and the identity map I in a *residual* one, and C is an output function set by the task: the identity for regression, with one output or several, the logistic sigmoid of each output for multilabel classification, and the softmax for multiclass classification (Bishop, 2006, Sections 5.1 and 5.2). So the hidden states are $h_0 = XE$ and $h_\ell = B_\ell(h_{\ell-1})$ in a skipless network and $h_\ell = h_{\ell-1} + B_\ell(h_{\ell-1})$ in a residual one, the readout gives $y = h_L D$, and

Proven	**ReLU**	**GELU**	**tanh**	**ReLU^2, ReGLU**	**SwiGLU, GeGLU**	**Attention**
Can a skipless network equal a residual one? Without normalization:						
one block, same width	measure zero Thm. 3.1(i)		never Thm. 3.3			never Thm. 3.4
one block, more hidden units or heads	exactly d_{model} more Thm. 3.1(ii)		never Thm. 3.3			never Thm. 3.4
networks of one block, same width	measure zero Prop. 3.2		never Thm. 4.1(ii)	never Thm. 4.1(i)		never Thm. A.13(iii)
deeper networks, same depth and widths	measure zero if the first block has fewer than $2d_{\text{model}}$ units, else unknown Thm. 4.2	measure zero companion	never Thm. 4.1(ii)	never Thm. 4.1(i)		measure zero companion
networks, more hidden units or heads per block	for almost all weights, d_{model} more suffice Prop. 4.3, and the first block needs $2d_{\text{model}}$ Thm. 4.2; every block needs $2d_{\text{model}}$ if contractive Prop. 4.4		never Thm. 4.1(ii)	never Thm. 4.1(i)		unknown
With pre-RMSNorm:						
networks, $\varepsilon > 0$	measure zero, at any width Thm. 4.5			never Thm. 4.1(i)		measure zero, at any width Thm. 4.5
networks, $\varepsilon = 0$	measure zero, at any width Thm. 4.5					
Is every skipless network a limit of residual ones of the same widths?						
without normalization	yes Prop. 5.1(i)					
with pre-RMSNorm, $\varepsilon = 0$	yes Prop. 5.1(ii)					
with pre-RMSNorm, $\varepsilon > 0$	unknown					

Table 1: What is proven, by setting. Never: for no weights. Measure zero: only for a null set of weights. Exactly d_{model} more: for almost every block, d_{model} more hidden units suffice and fewer do not. Contractive: as in Proposition 4.4. Companion: Karbevski and Mijoski (2026). No biases. Scope in Section 2.

$\mathcal{M}(X) = C(y)$. The function C has no weights, and we ignore it: equality of networks, and every result below, concern y. Networks with the same y have the same $\mathcal{M}$, but not conversely; a softmax, for instance, does not change when the same number is added to every output. In a *pre-RMSNorm* network every block and the readout read the normalized state $N(h)$, while the skips carry h: in the display, B_ℓ and D become $B_\ell \circ N$ and $D \circ N$, so the skipless and the residual network compute $h_\ell = B_\ell\big(N(h_{\ell-1})\big)$ and $h_\ell = h_{\ell-1} + B_\ell\big(N(h_{\ell-1})\big)$, and $y = N(h_L)D$. Here N divides each row v of h by $\sqrt{\|v\|^2/d_{\text{model}} + \varepsilon}$, with $\varepsilon \geq 0$. For $\varepsilon = 0$, N is defined on hidden states without zero rows, a network is defined at the inputs at which no state that it normalizes has a zero row, and two networks are equal if they agree on a dense set of inputs at which both are defined.

No biases. Blocks have no biases, and the answers depend on it. With biases, a gated unit whose gate read is zero has a constant gate, so it is affine in x, and d_{model} such units absorb a skip. Appendix C shows that, with biases, every ReLU, GELU, ReLU^2 and gated block absorbs its skip, and no tanh or attention block does; almost every ReLU, GELU and gated block needs exactly d_{model} more units. The gate matters even without biases: the logistic sigmoid gate of the original GLU does not vanish at 0, so a unit with a zero gate read is linear, and d_{model} such units absorb a skip.

Scope. Every MLP block has at least d_{model} hidden units, so that its read can have rank d_{model}. Residual networks have $ED \neq 0$. For simplicity, all MLP blocks of a network are of one kind, all attention blocks of a network have the same number of heads, and $d_{\text{in}}, d_{\text{out}} \geq d_{\text{model}} \geq 2$. Most

results hold more generally, and the appendix states them so. A statement holds for *generic* weights, or for almost every choice of weights, if it holds outside a set of Lebesgue measure zero.

3 One block

For almost every ReLU or GELU block, a skip is worth exactly d_{model} hidden units. For ReLU^2, gated, $\tanh$ *and attention blocks it is worth more than any number.*

3.1 ReLU and GELU blocks: a skip is worth d_{model} units

Both activations satisfy

$$\sigma(z) - \sigma(-z) = z, \tag{3}$$

so a unit and its mirror image, with the opposite read and the opposite write, add up to a linear term. This gives two ways to absorb xM.

With more units. Let unit i have read w_i, a column of W_{read}, and write $\hat{w}_i$, a row of W_{write}. For a row $\gamma \in \mathbb{R}^{1\times d_{\text{model}}}$, a new unit with read $-w_i$ and write $-\gamma$, with the write of unit i changed to $\hat{w}_i + \gamma$, adds $(xw_i)\,\gamma$ to the block, by (3). So each new unit adds one rank-one term, and d_{model} new units, mirroring d_{model} units whose reads span $\mathbb{R}^{d_{\text{model}}}$, add any xM.

At the same width. Flipping the read of unit i to $-w_i$ changes the block by $-(xw_i)\,\hat{w}_i$. So B_θ absorbs xM at the same width if some set S of units has $W_{\text{read}}[:,S]\,W_{\text{write}}[S,:] = -M$, where $W_{\text{read}}[:,S]$ and $W_{\text{write}}[S,:]$ keep the columns and rows of the units in S.

The next theorem says that at the same width these flips are the only way, and that fewer than d_{model} new units do not suffice. A block is *nondegenerate* if the columns of W_{read} are nonzero and pairwise non-collinear and the rows of W_{write} are nonzero.

Theorem 3.1 (ReLU and GELU blocks)**.** *Let σ be ReLU or GELU, and fix a matrix $M \neq 0$.*
(i) *If B_θ is nondegenerate, some block $B_{\theta'}$ of the same width satisfies (2) if and only if some set S of hidden units has $W_{\text{read}}[:,S]\,W_{\text{write}}[S,:] = -M$. For generic θ, no block of the same width does.*
(ii) *If M is invertible and W_{read} has rank d_{model}, some block $B_{\theta'}$ with d_{model} more hidden units satisfies (2), and for generic θ none with fewer does.*

Proof in Appendix A.1.

In (i), for each of the finitely many sets S, the equation $W_{\text{read}}[:,S]\,W_{\text{write}}[S,:] = -M$ fails when the writes vanish, so it holds only on a set of weights of measure zero. By (ii), for almost every block the skip, $M = I$, is worth exactly d_{model} units.

The proof matches the units of the two blocks. The even part of the activation, $\big(\sigma(z)+\sigma(-z)\big)/2$, matches each unit of the absorbing block to a unit of B_θ, up to sign and scale: for ReLU through the kinks of $|z|$, and for GELU through the linear independence of ridge functions, which depend on x only through one product xw. The odd part, $z/2$, turns each sign flip into a linear term, as Petzka et al. (2020, Section 3) show for two-layer ReLU networks with one output. Each extra unit then adds at most one rank-one term, and $M = I$ has rank d_{model}. Identity (3) alone is not enough. The activation $z/2 + |z|^3$ satisfies it, but scaling the read of a unit by $c > 0$ and its write by c^{-3} keeps the even part of the unit and changes its linear part, which gives more ways to absorb xM (Remark A.8).

Networks of one block. A residual network of one block is a block with the reads EW_{read} and the writes $W_{\text{write}}D$, plus the linear term XED: it computes $X \mapsto XED + f(XEW_{\text{read}})\,W_{\text{write}}D$. The proof of Theorem 3.1(i) holds for such rectangular reads, writes and linear terms, which gives the following.

Proposition 3.2 (Networks of one block)**.** *Let σ be ReLU or GELU. For almost every choice of weights, a residual network of one block equals no skipless network of one block of the same width.*

Proof in Appendix A.1.

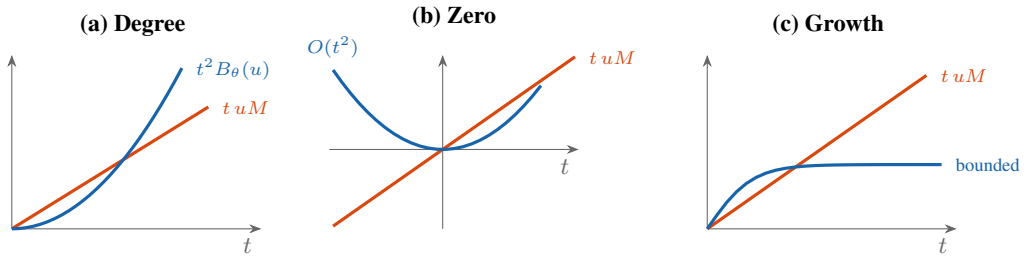


Figure 1: Why no width of ReLU^2, gated or tanh units absorbs the skip. Along a ray $x = tu$ the skip grows like t. (a) A ReLU^2 or ReGLU block grows like t^2. (b) A SwiGLU or GeGLU block is of order t^2 near $t = 0$. (c) A tanh block stays bounded.

3.2 ReLU^2, gated and tanh blocks

Recall that a block absorbs xM if a block of the same kind satisfies (2): $xM + B_\theta(x) = B_{\theta'}(x)$ for all x.

Theorem 3.3 (No width suffices)**.** *Let B_θ be a* tanh*, ReLU^2, ReGLU, SwiGLU or GeGLU block, or more generally a block that is positively homogeneous of degree $k \neq 1$, or a gated block whose gate vanishes at 0 and is differentiable there. For every $M \neq 0$, no block $B_{\theta'}$ of the same kind and of any width satisfies (2).*

Proof in Appendix A.1.

The three cases of Figure 1 are three one-line proofs. A homogeneous block satisfies $B_\theta(tu) = t^k B_\theta(u)$ for $t > 0$, so along a ray (2) reads $t\,uM = t^k\big(B_{\theta'}(u) - B_\theta(u)\big)$, which forces $uM = 0$ for every u when $k \neq 1$. A gate that vanishes at 0 makes every unit of order t^2 near the origin, so both blocks have derivative 0 there, while xM has derivative M. A tanh block is bounded, and xM is not. The condition $k \neq 1$ is needed: a linear block $x \mapsto xA$, whose activation is the identity, absorbs any skip, since $xM + xA = x(M + A)$.

3.3 Attention blocks

An *attention block* with parameters $\theta = (W_{Q,i}, W_{K,i}, W_{V,i}, W_{O,i})_{i=1}^H$ maps $h \in \mathbb{R}^{T \times d_{\text{model}}}$, whose rows are the $T \geq 2$ tokens, to

$$B_\theta(h) = \sum_{i=1}^{H} P_{\Gamma_i}(h)\, h\, W_{V,i} W_{O,i}, \qquad P_\Gamma(h) = \operatorname{softmax}\big(h\,\Gamma\, h^\top\big), \qquad \Gamma_i = W_{Q,i} W_{K,i}^\top / \sqrt{d_h}, \tag{4}$$

where $W_{Q,i}, W_{K,i}, W_{V,i} \in \mathbb{R}^{d_{\text{model}} \times d_h}$ and $W_{O,i} \in \mathbb{R}^{d_h \times d_{\text{model}}}$, with d_h the head dimension. The softmax acts on each row of the $T \times T$ matrix of scores, with or without a causal mask, so each row of $P_\Gamma(h)$ is a probability vector. The read of the block holds the matrices $W_{Q,i}$, $W_{K,i}$ and $W_{V,i}$, and its write the matrices $W_{O,i}$. Head i has the score matrix Γ_i and the value–output product $W_{V,i} W_{O,i}$.

Attention mixes tokens, and a skip keeps each token where it is. Suppose that $hM + B_\theta(h) = B_{\theta'}(h)$ for all h; two inputs show that $M = 0$. If all tokens are equal, mixing changes nothing, so the sums of the value–output products over the heads of the two blocks must differ by M. Now let one token be zero, placed last when the mask is causal. Its query is zero, so every head gives it the average of the tokens it sees, and the two blocks differ there by this mean times M. The skip adds zero to this token, so this mean times M vanishes for every choice of the other tokens, and $M = 0$.

Theorem 3.4 (Attention blocks)**.** *Let θ and θ' be parameters of attention blocks with any numbers of heads and any head dimensions, and let $M \neq 0$ be a $d_{\text{model}} \times d_{\text{model}}$ matrix. For almost every $h \in \mathbb{R}^{T \times d_{\text{model}}}$, $hM + B_\theta(h) \neq B_{\theta'}(h)$.*

A special case of Theorem A.13, proved in Appendix A.2.

The same two inputs show that a residual network of one attention block equals no skipless network of one attention block, with any number of heads (Theorem A.13(iii)). The proof also bounds

approximate absorption: on some input with rows of norm at most 1, one row of $hM + B_\theta(h) - B_{\theta'}(h)$ has norm at least $\|M\|/3$, with $\|M\|$ the spectral norm (Theorem A.13(ii)). The proof uses only that each row of attention weights sums to 1 and is uniform for a zero query, so it also holds with rotary embeddings. With three or more tokens, the function of an attention block determines the score matrices and the value–output products of its heads, up to merging heads that share a score matrix and dropping heads that output zero (Proposition A.15 in Appendix A.2).

4 Networks

Between an embedding and a readout, the picture survives. Without normalization, no residual network of ReLU^2, gated or $\tanh$ *blocks equals a skipless network, and a skipless ReLU or GELU network needs a wide first block. Under RMSNorm, almost no residual network equals a skipless one.*

Recall that a network embeds its input as $h_0 = XE$ and reads out $y = h_L D$, and that its skips add $h_{\ell-1}$ at every block.

Theorem 4.1 (Networks that never absorb their skips)**.** *Let the residual network have $ED \neq 0$.*
(i) *Without normalization or with pre-RMSNorm and $\varepsilon > 0$, let the blocks be gated blocks whose gate vanishes at 0 and is differentiable there, or blocks that are positively homogeneous of degree $k > 1$. No skipless network of blocks of the same kind, of any depth and widths, equals the residual network.*
(ii) *Without normalization, no skipless network of* $\tanh$ *blocks, of any depth and widths, equals a residual network of* $\tanh$ *blocks.*

Proof in Appendix A.1.

With $E = D = I$, this covers compositions of blocks. The reasons are those of Figure 1, now for the whole network. A gated block, and a homogeneous block of degree $k > 1$, has derivative 0 at the origin. So every skipless network of them has Jacobian 0 at the input $X = 0$, while a residual network has Jacobian $ED \neq 0$ there, or $ED/\sqrt{\varepsilon}$ under RMSNorm. This comparison of derivatives needs $k > 1$. A skipless $\tanh$ network is bounded, while a residual one grows like XED along rays.

ReLU and GELU networks need a wide first block.

Theorem 4.2 (ReLU and GELU networks)**.** *Consider networks of ReLU blocks, or of GELU blocks, without normalization. For almost every choice of weights, a residual network equals no skipless network whose first block has fewer than $2d_{\text{model}}$ hidden units, whatever its depth and its other widths.*

Proof in Appendix A.1.

For ReLU the proof compares ranks. Near almost every input, a generic residual ReLU network is a linear map of rank d_{model}. No unit of the first block is active at both an input and its negative, so with fewer than $2d_{\text{model}}$ units a skipless network has fewer than d_{model} active units at one of the two, and some direction of inputs leaves it unchanged there. GELU reduces to ReLU along rays: $|\text{GELU}(tz) - t\,\text{ReLU}(z)| < 0.17$ for all z and all $t > 0$, so as $t \to \infty$ the output of a GELU network at the input tX, divided by t, tends to the output at X of the ReLU network with the same weights. Two GELU networks that are equal therefore stay equal with ReLU in place of GELU.

Conversely, the construction of Section 3.1 works block by block.

Proposition 4.3 (Extra units)**.** *Let σ be ReLU or GELU. Without normalization, a residual network whose reads all have rank d_{model} equals a skipless network with d_{model} more hidden units in every block.*

Proof in Appendix A.1.

So d_{model} extra units per block suffice, and the first block of a skipless network needs at least $2d_{\text{model}}$ units (Theorem 4.2). For the first block, the two bounds meet when the residual first block has d_{model} units. A block B_ℓ is a *contraction* if $\|B_\ell(h) - B_\ell(h')\| < \|h - h'\|$ for all tokens $h \neq h'$, in the Euclidean norm. As ReLU is 1-Lipschitz, a ReLU block with $\|W_{\text{read}}\|\,\|W_{\text{write}}\| < 1$, in spectral norm, is a contraction.

Proposition 4.4 (Contractive blocks)**.** *Let σ be ReLU or GELU, without normalization, and let E and D have rank d_{model}. If every block of a residual network has $\|W_{\text{read}}\| \, \|W_{\text{write}}\| < 1$, every skipless network of σ blocks with the same d_{model} that equals it, of any depth, has at least $2d_{\text{model}}$ hidden units in every block.*

Proof in Appendix A.1.

This condition holds on an open set of weights, which contains every network whose writes are small enough for fixed reads, E and D. When every block of the residual network also has d_{model} hidden units and an invertible read, an equal skipless network of the same depth needs exactly d_{model} extra units in every block, by Proposition 4.3. The proof turns on injectivity. A residual block whose ReLU block is a contraction maps distinct tokens to distinct tokens. So every block of an equal skipless network must do so too, and a ReLU block without biases does so only with $2d_{\text{model}}$ units, since no unit is active at both h and $-h$. For ReLU networks the condition cannot simply be dropped. On another open set of weights, the first block maps each token h to nearly $(|h_1|, \dots, |h_{d_{\text{model}}}|)$, the units of later blocks are never active, and d_{model} extra units in the first block suffice at every depth (Remark A.12). For GELU networks without the condition, the question is open.

RMSNorm: the skip keeps the norm of each token. A skipless network sees its input only through its first normalization, and $N(ch)$ depends on $c > 0$ only through ε/c^2: with $\varepsilon = 0$ the norm of each token is lost, and with $\varepsilon > 0$ its effect fades like ε/c^2 as c grows.

Theorem 4.5 (Networks under RMSNorm)**.** *Consider pre-RMSNorm networks, with $\varepsilon \geq 0$ and blocks of any kind and width. For almost every choice of weights, a residual network equals no skipless network, whatever its depth, widths and kinds of blocks.*

Proof in Appendix A.3.

Scale an input X by a factor $t \to \infty$. A skipless network with embedding E' depends on t only through $N(tXE')$, so its output converges at rate $1/t^2$ or faster. In a residual network the skips carry tXE to the last normalization, and every block reads, up to $O(1/t)$, the same normalized input and adds a bounded output. These outputs turn each token by an angle of order $1/t$, so the output of the network converges only at rate $1/t$, unless in every token they add up to a multiple of the token itself. For almost every choice of weights, at some input they do not.

5 Skipless networks are limits of residual ones

A skip removes no function in the limit: without normalization, residual networks of the same widths approach every skipless network.

Proposition 5.1 (Skipless networks as limits of residual networks)**.** *Let F be a skipless network whose blocks are locally Lipschitz.*
(i) Without normalization. *The network F is a limit of residual networks with blocks of the same kinds and widths, uniformly on bounded sets of inputs.*
(ii) Under pre-RMSNorm with $\varepsilon = 0$. *The network F is a limit of residual networks with blocks of the same kinds and widths, uniformly on compact sets of inputs at which F is defined.*

Proof in Appendix A.4.

Blocks that outweigh their skips. The proof scales the reads of block ℓ by $t^{1-\ell}$, its writes by t^{ℓ} and the readout by t^{-L}. Block ℓ then sees the hidden state of F, up to $O(1/t)$, and outweighs its skip by a factor t, so the residual network computes $F + O(1/t)$ (Remark A.16). The large constants sit in the writes, so no activation sees a large argument, and the construction applies alike to $\tanh$, GELU, ReLU, ReLU^2, gated and attention blocks. The weights diverge as $t \to \infty$: the proposition says which functions residual networks reach in the limit, and leaves open how to parametrize them for training.

Under RMSNorm. With $\varepsilon = 0$, RMSNorm ignores the size of the hidden states, so only the writes grow. With $\varepsilon > 0$ the construction fails, since on a hidden state of size t^{ℓ} RMSNorm acts as RMSNorm with $\varepsilon/t^{2\ell}$; whether skipless networks are limits of residual ones there is open.

6 Related work

Expressivity of residual networks. Lin and Jegelka (2018) show that residual ReLU networks with one hidden unit per block are universal approximators, which networks of width d_{in} without skips are not; they count the identity map as d_{model} hidden units. For blocks without biases and with at least d_{model} units, Theorem 3.1(ii) shows that this count is exact for almost every ReLU or GELU block, and Theorem 3.3 that no count works for ReLU^2, gated and $\tanh$ blocks. With biases, as in their blocks, the count stays exact for almost every ReLU or GELU block with at least d_{model} units (Theorem C.2 in Appendix C). Hardt and Ma (2017) show that deep linear residual networks have no spurious local optima, and that residual ReLU networks can fit any finite sample once they have more parameters than samples. These results concern approximation, optimization or finite samples; ours concern exact equality of functions, block by block and activation by activation.

Identifiability. For $\tanh$ networks, the function determines the weights up to sign flips and permutations of hidden units (Fefferman, 1994; Sussmann, 1992), and similar statements hold for other activations (Vlačić and Bölcskei, 2022) and for ReLU networks (Petzka et al., 2020; Phuong and Lampert, 2020; Rolnick and Kording, 2020). Our proofs identify the units of a block from its function in the same spirit, and then ask what is left for the linear term of the skip.

Training without skips. Skipless networks train with careful initialization or activation shaping (Martens et al., 2021; Zhang et al., 2022), and skipless transformers with modified attention (He et al., 2023; Noci et al., 2023) or simplified blocks (He and Hofmann, 2024). Ji et al. (2026) train a skipless ViT-B that outperforms the residual one. Our results say when a skipless block or network can compute exactly what a residual one computes, and with how many units.

7 Conclusion

Can a block absorb its skip? Without biases, it depends on the activation. At the same width, almost no block absorbs its skip. Almost every ReLU or GELU block needs exactly d_{model} more units, and ReLU^2, gated, $\tanh$ and attention blocks cannot absorb the skip at any width. Between an embedding and a readout and without normalization, no skipless network of ReLU^2, gated or $\tanh$ blocks equals a residual one, and a skipless ReLU or GELU network needs a first block of at least $2d_{\text{model}}$ units to equal almost any residual one. Under RMSNorm, with any $\varepsilon \geq 0$, no width suffices for almost every residual network. So, without normalization or biases, a trained residual ReLU or GELU network whose reads have rank d_{model} can be rewritten exactly as skipless networks with d_{model} more units per block, and a ReLU^2, gated or $\tanh$ network with $ED \neq 0$ cannot be rewritten at any width. With biases, ReLU^2 and gated networks can be rewritten too (Appendix C).

Open questions. The unknown cells and the conditions of Table 1 leave three questions, and a fourth lies beyond the table.

(1) Networks of attention blocks at depth, without normalization. At the same widths, a companion paper answers this by counting functional dimensions, the numbers of directions in which the parameters move the function (Karbevski and Mijoski, 2026); with more heads it is open.

(2) How many extra units ReLU and GELU networks need, without normalization, when the blocks are wider than d_{model}, have reads of rank below d_{model}, or are not contractions. A skipless GELU network needs at least $Ld_{\text{model}}/2$ extra units in total (Karbevski and Mijoski, 2026).

(3) Skipless networks as limits of residual networks under RMSNorm with $\varepsilon > 0$.

(4) Approximate absorption: how many units a skip is worth at a given error.

Use of AI

The conception of the study is the authors' own. One author obtained their results, proofs and experiments working with AI assistants, through continuous conversation; the assistants were used to find, verify and simplify proofs and to write the code of the experiments, which the authors validated.

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A Proofs

A.1 MLP blocks and networks (Sections 3 and 4)

We recall Theorems 3.1 and 3.3 and Proposition 3.2 from Section 3 and Theorems 4.1 and 4.2 and Propositions 4.3 and 4.4 from Section 4, for the question (2). Below we write B for B_θ, and B' for $B_{\theta'}$ with $\theta' = (W'_{\text{read}}, W'_{\text{write}})$.

Theorem 3.1 (ReLU and GELU blocks)**.** *Let σ be ReLU or GELU, and fix a matrix $M \neq 0$.*
(i) *If B_θ is nondegenerate, some block $B_{\theta'}$ of the same width satisfies (2) if and only if some set S of hidden units has $W_{\text{read}}[:, S]\, W_{\text{write}}[S, :] = -M$. For generic θ, no block of the same width does.*
(ii) *If M is invertible and W_{read} has rank d_{model}, some block $B_{\theta'}$ with d_{model} more hidden units satisfies (2), and for generic θ none with fewer does.*

Proposition 3.2 (Networks of one block)**.** *Let σ be ReLU or GELU. For almost every choice of weights, a residual network of one block equals no skipless network of one block of the same width.*

Theorem 3.3 (No width suffices)**.** *Let B_θ be a $\tanh$, ReLU^2, ReGLU, SwiGLU or GeGLU block, or more generally a block that is positively homogeneous of degree $k \neq 1$, or a gated block whose gate vanishes at 0 and is differentiable there. For every $M \neq 0$, no block $B_{\theta'}$ of the same kind and of any width satisfies (2).*

Theorem 4.1 (Networks that never absorb their skips)**.** *Let the residual network have $ED \neq 0$.*
(i) *Without normalization or with pre-RMSNorm and $\varepsilon > 0$, let the blocks be gated blocks whose gate vanishes at 0 and is differentiable there, or blocks that are positively homogeneous of degree $k > 1$. No skipless network of blocks of the same kind, of any depth and widths, equals the residual network.*
(ii) *Without normalization, no skipless network of $\tanh$ blocks, of any depth and widths, equals a residual network of $\tanh$ blocks.*

Theorem 4.2 (ReLU and GELU networks)**.** *Consider networks of ReLU blocks, or of GELU blocks, without normalization. For almost every choice of weights, a residual network equals no skipless network whose first block has fewer than $2d_{\text{model}}$ hidden units, whatever its depth and its other widths.*

Proposition 4.3 (Extra units)**.** *Let σ be ReLU or GELU. Without normalization, a residual network whose reads all have rank d_{model} equals a skipless network with d_{model} more hidden units in every block.*

Proposition 4.4 (Contractive blocks)**.** *Let σ be ReLU or GELU, without normalization, and let E and D have rank d_{model}. If every block of a residual network has $\|W_{\text{read}}\|\,\|W_{\text{write}}\| < 1$, every skipless network of σ blocks with the same d_{model} that equals it, of any depth, has at least $2d_{\text{model}}$ hidden units in every block.*

Overview. Theorem 3.3 compares orders. Along rays from the origin a linear skip scales like λ and a block that is homogeneous of degree $k \neq 1$ like λ^k, and near the origin a gated block is of second order, so the two sides of the absorption equation cannot agree (Propositions A.2 and A.3); comparing Jacobians at the origin extends this to networks of any depth, Theorem 4.1 (Corollary A.4). Theorem 3.1(i) is the substantial case. The absorption equation splits into its odd and even parts in x. The even part determines the hidden units of the absorbing block up to sign, and for ReLU up to a positive rescaling, through the kinks for ReLU (Theorem A.5) and through linear independence of ridge functions for GELU (Lemma A.6 and Theorem A.7). The odd part of ReLU and GELU is exactly $z/2$, so the odd part of the equation is linear and reads off the set S of units whose signs flip; conversely, flipping the signs of the units in S absorbs the skip whenever the condition holds. The generic clause holds because the condition is a finite union of algebraic sets of positive codimension (Corollary A.10), and Remark A.9 covers singular M. Theorem 3.1(ii) counts the extra units that the matching requires (Proposition A.11). Proposition 4.3 applies Theorem 3.1(ii) block by block. Theorem 4.2 compares the ranks of the linear pieces of a ReLU network at an input and at its negative, and reduces GELU to ReLU along rays.

The arguments show that the function of a block determines its parameters, in the manner used for $\tanh$ networks by Sussmann (1992) and Fefferman (1994) and for ReLU networks by Phuong and Lampert (2020), Petzka et al. (2020) and Rolnick and Kording (2020): kink alignment for ReLU, ridge-function independence for GELU. For two-layer ReLU networks with a single output, Petzka et al. (2020, Section 3) already give the corresponding sign flips, show that irreducible networks computing the same function differ only by these flips, permutations and positive rescalings (their Theorem 1), and state the matching criterion for removing a linear term. Lemma A.1 reduces any invertible skip to the identity skip.

Setting. Throughout, $W_{\text{read}} \in \mathbb{R}^{d_{\text{model}} \times m}$ and $W_{\text{write}} \in \mathbb{R}^{m \times d_{\text{model}}}$, σ acts coordinatewise, w_i is column i of W_{read} and $\hat{w}_i$ is row i of W_{write}, and inputs x are row vectors. For $S \subseteq \{1, \dots, m\}$, $W_{\text{read}}[:, S]$ is the submatrix of columns in S, $W_{\text{write}}[S, :]$ the submatrix of rows in S, and Π_S the diagonal projector with $(\Pi_S)_{ii} = 1$ exactly when $i \in S$, so that $W_{\text{read}} \Pi_S W_{\text{write}} = W_{\text{read}}[:, S] W_{\text{write}}[S, :]$. A family of vectors is *pairwise non-collinear* if no one of them is a scalar multiple of another, and a block is *nondegenerate* if the columns of W_{read} are nonzero and pairwise non-collinear and the rows of W_{write} are nonzero. The question is whether there is a block $B'(x) = \sigma(xW'_{\text{read}})W'_{\text{write}}$ of the same width, with $W'_{\text{read}} \in \mathbb{R}^{d_{\text{model}} \times m}$ and $W'_{\text{write}} \in \mathbb{R}^{m \times d_{\text{model}}}$, such that

$$x + \sigma(xW_{\text{read}})W_{\text{write}} = \sigma(xW'_{\text{read}})W'_{\text{write}} \qquad \text{for all } x \in \mathbb{R}^{1 \times d_{\text{model}}}. \tag{5}$$

Lemma A.1 (Invertible reduction)**.** *Let $M \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ be invertible. Then*

$$xM + \sigma(xW_{\text{read}})W_{\text{write}} = \sigma(xW'_{\text{read}})W'_{\text{write}} \quad \textit{for all } x$$

if and only if

$$x + \sigma(xW_{\text{read}})W_{\text{write}}M^{-1} = \sigma(xW'_{\text{read}})W'_{\text{write}}M^{-1} \quad \textit{for all } x.$$

Proof. Right-multiply by M^{-1}, which is invertible. □

Any invertible linear skip therefore reduces to the identity skip.

Proposition A.2 (Homogeneous blocks)**.** *Let B and B' be blocks that are positively homogeneous of degree $k \neq 1$, of any widths, and let $M \neq 0$. Then $xM + B(x) = B'(x)$ fails for some x.*

Examples are ReLU^2 and ReGLU blocks without biases, both of degree 2.

Proof. Outline. We compare the two sides along rays through the origin: a term of degree one and a term of degree $k \neq 1$ cannot agree at every scale.

Suppose it holds for all x and evaluate at λx with $\lambda > 0$: $\lambda xM + \lambda^k B(x) = \lambda^k B'(x)$. Dividing by λ gives $xM = \lambda^{k-1}\big(B'(x) - B(x)\big)$, and $\lambda^{k-1} \to 0$ as $\lambda \to 0$ if $k > 1$ and as $\lambda \to \infty$ if $k < 1$. So $xM = 0$ for all x, which contradicts $M \neq 0$. □

Proposition A.3 (Gated blocks)**.** *Let B and B' be gated blocks without biases, of any widths, whose gates are differentiable at 0 and vanish there, and let $M \neq 0$. Then $xM + B(x) = B'(x)$ fails for some x.*

Examples are SwiGLU and GeGLU blocks, which Shazeer (2020) defines without biases.

Proof. Outline. Near the origin a gated block is of second order in x, while xM is of first order.

A unit of a gated block computes $g(u \cdot x)\,(v \cdot x)$. Since $g(0) = 0$ and g is differentiable at 0, $|g(u \cdot x)| \leq c\,\|x\|$ for small $\|x\|$, so the unit is $O(\|x\|^2)$, and so are B and B'. If $xM + B(x) = B'(x)$ for all x, then $xM = O(\|x\|^2)$ as $x \to 0$. Along a ray $x = tz$ this reads $t\,zM = O(t^2)$, so $zM = 0$ for every z, which contradicts $M \neq 0$. □

Corollary A.4 (Any depth)**.** *Let every block be as in Proposition A.2 with $k > 1$ or as in Proposition A.3, of any widths. Let F be a composition of residual blocks $h \mapsto h + B_\ell(h)$ and H a composition of one or more skipless blocks. Then $F \neq H$. With embeddings and readouts, the residual network $X \mapsto F(XE)D$ differs from the skipless network $X \mapsto H(XE')D'$ whenever $ED \neq 0$, also with pre-RMSNorm and $\varepsilon > 0$. Without normalization the same holds for $\tanh$ blocks.*

Proof. Each such block maps 0 to 0 and has Jacobian 0 there, because it is $O(\|x\|^k)$ with $k > 1$, respectively $O(\|x\|^2)$. By the chain rule F has Jacobian I at the origin and H has Jacobian 0; with embeddings and readouts these become ED and 0. Under pre-RMSNorm with $\varepsilon > 0$, $N(0) = 0$ and N has Jacobian $I/\sqrt{\varepsilon}$ at 0, so every block, read through N, still has Jacobian 0 there, and the residual and the skipless network have Jacobians $ED/\sqrt{\varepsilon}$ and 0 at the origin. For $\tanh$ blocks without normalization, H and $F(h) - h$ are bounded, so $F(XE)D - H(XE')D'$ differs from XED by a bounded function and does not vanish when $ED \neq 0$. □

Parity. The remaining activations have an odd part that is exactly linear. For ReLU, $\mathrm{ReLU}(z) = |z|/2 + z/2$. For GELU, $\mathrm{GELU}(z) = z\Phi(z)$ with Φ the standard normal distribution function, and since $\Phi(-z) = 1 - \Phi(z)$,

$$\mathrm{GELU}(z) = \Psi(z) + z/2, \qquad \Psi(z) := z\big(\Phi(z) - \tfrac{1}{2}\big), \qquad \Psi(-z) = \Psi(z).$$

The even parts, $|z|/2$ and Ψ, match the hidden units of the two blocks; the odd part, $z/2$, turns a sign flip into a linear term and gives the absorption condition.

Theorem A.5 (Identity absorption for ReLU)**.** *Let $\sigma = \mathrm{ReLU}$, $d_{\mathrm{model}} \geq 2$, and let the block be nondegenerate. Then (5) has a solution if and only if*

$$W_{\mathrm{read}}[:, S]\, W_{\mathrm{write}}[S, :] = -I_{d_{\mathrm{model}}} \quad \textit{for some } S. \tag{6}$$

Proof. Outline. Sufficiency is a construction: flipping the signs of the hidden units indexed by S produces the missing identity term. For necessity we split (5) into its odd and even parts in x, use the kinks of the even part to match the hidden units of the two blocks one to one, and read the index set S off the signs of the matching.

Sufficiency. Let $F = I_m - 2\Pi_S$, a diagonal sign matrix, and set $W'_{\mathrm{read}} = W_{\mathrm{read}}F$, $W'_{\mathrm{write}} = W_{\mathrm{write}}$. Since $|xW_{\mathrm{read}}F| = |xW_{\mathrm{read}}|$ coordinatewise,

$$\mathrm{ReLU}(xW'_{\mathrm{read}})W'_{\mathrm{write}} = \tfrac{1}{2}\big(xW_{\mathrm{read}}F + |xW_{\mathrm{read}}|\big)W_{\mathrm{write}},$$
$$x + \mathrm{ReLU}(xW_{\mathrm{read}})W_{\mathrm{write}} = \tfrac{1}{2}\big(xW_{\mathrm{read}} + |xW_{\mathrm{read}}|\big)W_{\mathrm{write}} + x,$$

and the two agree if and only if $\frac{1}{2}xW_{\mathrm{read}}(F - I_m)W_{\mathrm{write}} = x$ for all x, that is $-W_{\mathrm{read}}\Pi_S W_{\mathrm{write}} = I_{d_{\mathrm{model}}}$, which is (6).

Necessity. Suppose (5) holds and split both sides into odd and even parts in x:

$$W'_{\mathrm{read}}W'_{\mathrm{write}} = W_{\mathrm{read}}W_{\mathrm{write}} + 2I_{d_{\mathrm{model}}}, \qquad |xW'_{\mathrm{read}}|\, W'_{\mathrm{write}} = |xW_{\mathrm{read}}|\, W_{\mathrm{write}} \ \text{ for all } x. \tag{7}$$

Step 1: alignment of kinks. The map $x \mapsto |xW_{\mathrm{read}}|\, W_{\mathrm{write}}$ fails to be differentiable exactly on the union of the hyperplanes $\{xw_i = 0\}$; these are m distinct hyperplanes, each carrying a genuine kink since $\hat{w}_i \neq 0$. By the second identity in (7) the map $x \mapsto |xW'_{\mathrm{read}}|\, W'_{\mathrm{write}}$ has the same non-differentiability locus. Each of the m kink hyperplanes $\{x : xw_i = 0\}$ of the target needs at least one unit of B' whose kink lies on it, and B' has only m units; so each hyperplane gets exactly one unit, no two units share a hyperplane, and no kinks cancel. After relabeling, $W'_{\mathrm{read}} = W_{\mathrm{read}}C$ with $C = \mathrm{diag}(c_1, \ldots, c_m)$, all $c_i \neq 0$. *Step 2: coefficients.* Substituting, $|xW_{\mathrm{read}}C| = |xW_{\mathrm{read}}|\, |C|$, so $|xW_{\mathrm{read}}|\, |C|\, W'_{\mathrm{write}} = |xW_{\mathrm{read}}|\, W_{\mathrm{write}}$ for all x. For $d_{\mathrm{model}} \geq 2$ and pairwise non-collinear columns, the functions $|xw_i|$ are linearly independent modulo linear functions, because their kink hyperplanes are distinct; hence $|C|\, W'_{\mathrm{write}} = W_{\mathrm{write}}$ and $W'_{\mathrm{write}} = |C|^{-1}W_{\mathrm{write}}$. *Step 3: signs.* The first identity in (7) becomes $W_{\mathrm{read}}C|C|^{-1}W_{\mathrm{write}} = W_{\mathrm{read}}W_{\mathrm{write}} + 2I_{d_{\mathrm{model}}}$. With $\Sigma = C|C|^{-1} = \mathrm{diag}(\mathrm{sign}\, c_i)$ this reads $W_{\mathrm{read}}(\Sigma - I_m)W_{\mathrm{write}} = 2I_{d_{\mathrm{model}}}$; the diagonal of $\Sigma - I_m$ has entries 0 or -2, so with $S = \{i : c_i < 0\}$ we get $W_{\mathrm{read}}\Pi_S W_{\mathrm{write}} = -I_{d_{\mathrm{model}}}$, and $|S| \geq d_{\mathrm{model}}$ by the rank of $I_{d_{\mathrm{model}}}$. □

The GELU case needs one classical fact about ridge functions, whose short proof we include.

Lemma A.6 (Vanishing sums of ridge functions)**.** *Let $a_1, \ldots, a_r \in \mathbb{R}^{d_{\mathrm{model}}}$ be nonzero and pairwise non-collinear, and let $g_1, \ldots, g_r : \mathbb{R} \to \mathbb{R}^k$ be $r - 1$ times continuously differentiable with $\sum_i g_i(a_i \cdot x) = 0$ for all $x \in \mathbb{R}^{d_{\mathrm{model}}}$. Then every g_i is a polynomial of degree at most $r - 2$.*

Proof. Fix i. For each $j \neq i$ choose $u_j \in \mathbb{R}^{d_{\text{model}}}$ with $u_j \cdot a_j = 0$ and $u_j \cdot a_i \neq 0$, which exists because a_i is not a multiple of a_j. The directional derivative ∂_{u_j} annihilates every function of $a_j \cdot x$ and maps a function of $a_k \cdot x$ to a function of $a_k \cdot x$. Applying the product of the commuting operators ∂_{u_j}, $j \neq i$, to the sum therefore leaves $\prod_{j \neq i}(u_j \cdot a_i)\, g_i^{(r-1)}(a_i \cdot x) = 0$. Since $a_i \neq 0$, $a_i \cdot x$ takes every real value, so $g_i^{(r-1)} = 0$. □

Theorem A.7 (Identity absorption for GELU)**.** *Let $\sigma = \mathrm{GELU}$ and let the block be nondegenerate. Then (5) has a solution if and only if (6) holds.*

Proof. Outline. The proof follows that of Theorem A.5. Linear independence of ridge functions replaces the kink argument, and one extra step shows that matched hidden units have the same scale.

Sufficiency. With $F = I_m - 2\Pi_S$, $W'_{\text{read}} = W_{\text{read}}F$ and $W'_{\text{write}} = W_{\text{write}}$, evenness of Ψ gives $\Psi(xW_{\text{read}}F) = \Psi(xW_{\text{read}})$, so

$$\begin{aligned}\mathrm{GELU}(xW'_{\text{read}})W'_{\text{write}} &= \Psi(xW_{\text{read}})W_{\text{write}} + \tfrac{1}{2}xW_{\text{read}}FW_{\text{write}} \\ &= \mathrm{GELU}(xW_{\text{read}})W_{\text{write}} - xW_{\text{read}}\Pi_S W_{\text{write}} \\ &= \mathrm{GELU}(xW_{\text{read}})W_{\text{write}} + x.\end{aligned}$$

Necessity. Splitting (5) into even and odd parts gives

$$\Psi(xW'_{\text{read}})W'_{\text{write}} = \Psi(xW_{\text{read}})W_{\text{write}} \ \text{ for all } x, \qquad W'_{\text{read}}W'_{\text{write}} = W_{\text{read}}W_{\text{write}} + 2I_{d_{\text{model}}}. \tag{8}$$

Step 1: alignment of directions. Ψ is analytic and not a polynomial. Zero columns of W'_{read} contribute $\Psi(0) = 0$; group the other columns of W'_{read} and the columns of W_{read} into classes of collinear vectors. In the first identity of (8) the terms of each class form one ridge function in the direction of the class, and these ridge functions sum to zero, so each is a polynomial by Lemma A.6. The class of a column w_i of W_{read} contains no other column of W_{read}, so if it contained no column of W'_{read} its ridge function would be $-\Psi(xw_i)\,\hat{w}_i$, which is not a polynomial because $\hat{w}_i \neq 0$. Hence each of the m classes of the columns of W_{read} contains a nonzero column of W'_{read}, and since W'_{read} has only m columns, after relabeling $W'_{\text{read}} = W_{\text{read}}C$ with $C = \mathrm{diag}(c_i)$, $c_i \neq 0$. *Step 2: unit scale.* Collecting the terms of direction w_i on both sides gives, for each i, a ridge function $g_i(xw_i) := \Psi(|c_i|\,xw_i)\,(W'_{\text{write}})_{i,:} - \Psi(xw_i)\,(W_{\text{write}})_{i,:}$ with $\sum_i g_i(xw_i) = 0$. By Lemma A.6 each g_i is a polynomial. For $t \to +\infty$, $\Psi(t) = t/2 - t\,\bar{\Phi}(t)$ with $\bar{\Phi}(t) = 1 - \Phi(t)$ exponentially small, so $g_i(t) = \frac{t}{2}\big(|c_i|(W'_{\text{write}})_{i,:} - (W_{\text{write}})_{i,:}\big) + o(1)$; a polynomial with this behavior is linear, and it is even because Ψ is even, so it is constant, and letting $t \to 0$ shows the constant is zero. Hence $|c_i|(W'_{\text{write}})_{i,:} = (W_{\text{write}})_{i,:}$ and the exponentially small remainder $|c_i|\,t\,\bar{\Phi}(|c_i|t)\,(W'_{\text{write}})_{i,:} - t\,\bar{\Phi}(t)\,(W_{\text{write}})_{i,:}$ must vanish, which with $(W_{\text{write}})_{i,:} \neq 0$ forces $\bar{\Phi}(|c_i|t) = \bar{\Phi}(t)$ for all $t > 0$, so $|c_i| = 1$. Therefore $|C| = I_m$ and $W'_{\text{write}} = W_{\text{write}}$. *Step 3: signs.* Identical to the ReLU case: the second identity in (8) becomes $W_{\text{read}}(C - I_m)W_{\text{write}} = 2I_{d_{\text{model}}}$ with $C = \mathrm{diag}(\pm 1)$, so $S = \{i : c_i = -1\}$ satisfies (6) and $|S| \geq d_{\text{model}}$. □

Remark A.8 (The even part). Any activation whose even part ψ is analytic and not a polynomial matches the hidden units in the same way: Steps 1 and 2 of the proof of Theorem A.7 apply with ψ in place of Ψ (zero columns of W'_{read} now contribute a constant, which is a polynomial), except that the unit scale in Step 2 follows from a general fact: if $\sum_k a_k\psi(\alpha_k t)$ is a polynomial for distinct $\alpha_k > 0$, then every $a_k = 0$. Indeed, the Taylor coefficients $\psi_n \sum_k a_k\alpha_k^n$ vanish for all large n, infinitely many ψ_n are nonzero, and a generalized Vandermonde matrix with distinct positive nodes is invertible. Linear independence of the ridge functions alone does not suffice: for $\sigma(z) = z/2 + |z|^3$ the even part is homogeneous, so a unit can be rescaled together with its output row, and the unit scale is not forced. The odd part o of σ enters only through the flips: with the units matched, both directions reduce to $x = -2\sum_{i \in S} o(xw_i)\,\hat{w}_i$, with S the flipped units. For $o(z) = z/2$, that is $\sigma(z) - \sigma(-z) = z$, as for SiLU, whose even part is $\frac{z}{2}\tanh\frac{z}{2}$, this is (6), and $o(z) = \alpha z$ with $\alpha \neq 0$ only rescales it; a polynomial o of higher degree adds the condition that the units in S cancel its terms of degree three and higher, and a smooth o that is not a polynomial admits no S (Lemma A.6), so the skip is never absorbed at the same width.

Remark A.9 (Singular linear skips). Nothing in the proofs of Theorems A.5 and A.7 uses that the linear term is the identity. With xM in place of x on the left of (5), the odd part of the equation reads $W'_{\text{read}}W'_{\text{write}} = W_{\text{read}}W_{\text{write}} + 2M$, the even part is unchanged, and the sign step yields $W_{\text{read}}\Pi_S W_{\text{write}} = -M$. So for any $M \neq 0$, invertible or not, absorption holds if and only if $W_{\text{read}}[:, S]W_{\text{write}}[S, :] = -M$ for some S, with $|S| \geq \operatorname{rank} M$, which is a measure-zero condition on the weights for each fixed M. This gives Theorem 3.1(i) for singular M. The proofs also hold verbatim for rows $x \in \mathbb{R}^{1\times p}$ with $p \geq 2$, reads in $\mathbb{R}^{p\times m}$, writes in $\mathbb{R}^{m\times q}$ and $M \in \mathbb{R}^{p\times q}$: the kinks and the ridge functions live in $\mathbb{R}^p$, and the sign step gives $W_{\text{read}}[:, S]\, W_{\text{write}}[S, :] = -M$ for any shape. This covers networks of one block, with the reads EW_{read}, the writes $W_{\text{write}}D$ and $M = ED$, so that $p = d_{\text{in}}$ and $q = d_{\text{out}}$. For generic E, W_{read}, W_{write} and D, these reads and writes are nondegenerate, and $EW_{\text{read}}[:, S]\, W_{\text{write}}[S, :]\, D \neq -ED$ for every S, since the two sides differ at $W_{\text{write}} = 0$ when $ED \neq 0$.

Corollary A.10 (Generic disjointness for one block)**.** *Let σ be ReLU or GELU and $d_{\text{model}} \geq 2$. For generic weights,* (5) *has no solution.*

Proof. Outline. The condition of Theorems A.5 and A.7 is a finite union of algebraic sets of positive codimension, and such a union has measure zero.

For each of the finitely many S, (6) is d^2_{model} polynomial equations in the weights and defines a set of measure zero; the hypotheses of Theorems A.5 and A.7 hold outside another set of measure zero. So a generic residual block is not a skipless block of the same width, and, by the same argument with the two sides of (5) exchanged, a generic skipless block is not a residual block of the same width: neither class contains the other. □

Proposition A.11 (d_{model} extra units)**.** *Let σ be ReLU or GELU, let $d_{\text{model}} \geq 2$, and let $M \in \mathbb{R}^{d_{\text{model}}\times d_{\text{model}}}$ be invertible.*
(i) *If W_{read} has rank d_{model}, a block B' of width $m + d_{\text{model}}$ satisfies $xM + B(x) = B'(x)$ for all x.*
(ii) *For generic weights no block of width less than $m + d_{\text{model}}$ does.*

Proof. Outline. Two units with opposite reads produce a linear term of rank one, so d_{model} extra units produce xM. Conversely, the matching of the units in Theorems A.5 and A.7 shows that every extra unit adds at most one such term.

(i) As $\sigma(z) - \sigma(-z) = z$, a column w and rows a, b give $\sigma(xw)(a + b) + \sigma(-xw)(-b) = \sigma(xw)\, a + (xw)\, b$. Choose d_{model} units whose columns $w_{i_1}, \ldots, w_{i_{d_{\text{model}}}}$ form an invertible matrix W_K, and let $b_1, \ldots, b_{d_{\text{model}}}$ be the rows of $W_K^{-1}M$. Adding b_k to the row $\hat{w}_{i_k}$, and a unit with column $-w_{i_k}$ and row $-b_k$, for each k, gives a block of width $m + d_{\text{model}}$ that computes $\sigma(xW_{\text{read}})W_{\text{write}} + xW_K W_K^{-1}M$.

(ii) Outside a set of measure zero the block is nondegenerate, and for each of the finitely many sets S of units the determinant of $M + W_{\text{read}}[:, S]W_{\text{write}}[S, :]$ is nonzero, since it is a polynomial in the weights equal to $\det M \neq 0$ at $W_{\text{write}} = 0$. Let a block B' of width m', with columns w'_j and rows $\hat{w}'_j$, satisfy $xM + B(x) = B'(x)$. Its odd part gives $W'_{\text{read}}W'_{\text{write}} = W_{\text{read}}W_{\text{write}} + 2M$, and its even part $|xW'_{\text{read}}|\, W'_{\text{write}} = |xW_{\text{read}}|\, W_{\text{write}}$ for ReLU and $\Psi(xW'_{\text{read}})W'_{\text{write}} = \Psi(xW_{\text{read}})W_{\text{write}}$ for GELU. Group the nonzero columns w'_j and the columns w_i into classes of collinear vectors, write $w'_j = s_j u$ with a unit vector u of the class, $u = w_i/\|w_i\|$ in the class of w_i, and let $P_u = \|w_i\|\hat{w}_i$ there and $P_u = 0$ in a class without a column of W_{read}. The even part then forces, class by class, $\sum_j |s_j|\, \hat{w}'_j = P_u$ for ReLU, since the functions $|xu|$ of distinct classes are linearly independent (Step 2 of Theorem A.5), and for GELU, by Lemma A.6 and the independence of the functions $\Psi(\alpha t)$ for distinct $\alpha > 0$ modulo polynomials (Remark A.8), $\sum_{|s_j|=\alpha} \hat{w}'_j = \hat{w}_i$ if w_i lies in the class and $\alpha = \|w_i\|$, and 0 otherwise. With $Q_u = \sum_j s_j \hat{w}'_j$ the odd part reads $\sum_u u\,(Q_u - P_u) = 2M$. Call a class mixed if it contains w'_j of both signs, for GELU of both signs and equal $|s_j|$. A class that is not mixed has $Q_u = P_u$ or $Q_u = -P_u$; let S collect the units i whose class is not mixed and has $Q_u = -P_u \neq 0$. Then

$$2\big(M + W_{\text{read}}[:, S]W_{\text{write}}[S, :]\big) = \sum_{u \text{ mixed}} u\,(Q_u - P_u).$$

Every class of a column w_i contains some w'_j, since $P_u \neq 0$, and every mixed class contains at least two, so $m' \geq m$ plus the number of mixed classes. If $m' < m + d_{\text{model}}$, the right side has rank less than d_{model}, so the left side is singular, which the weights exclude. □

Proof of Theorems 3.3 and 4.1. For homogeneous and gated blocks, Theorem 3.3 is Propositions A.2 and A.3, and Theorem 4.1(i) is Corollary A.4. For $\tanh$ blocks, B_θ and $B_{\theta'}$ are bounded and xM is not, which gives Theorem 3.3, and Theorem 4.1(ii) is again Corollary A.4. □

Proof of Theorem 3.1. Part (i) is Theorems A.5 and A.7, with Lemma A.1 for invertible M and Remark A.9 for singular M; for each M the condition holds only on a set of weights of measure zero (Corollary A.10 and Remark A.9). Part (ii) is Proposition A.11. □

Proof of Proposition 3.2. A network of one block computes $X \mapsto XED + f(XEW_{\text{read}})\, W_{\text{write}} D$, and a skipless network of one block of the same width computes $X \mapsto f(XE'W'_{\text{read}})\, W'_{\text{write}} D'$. Both act on each row of X separately. By the rectangular case of Remark A.9, with the reads EW_{read}, the writes $W_{\text{write}}D$ and $M = ED$, they are equal only if $EW_{\text{read}}[:, S]\, W_{\text{write}}[S, :]\, D = -ED$ for some set S of units, or if the reads and writes are degenerate, and for generic weights neither happens. □

Proof of Theorem 4.2. MLP blocks act on each token separately, so we compare the networks on rows $x \in \mathbb{R}^{1\times d_{\text{in}}}$. Outside a set of weights of measure zero, E and D have rank d_{model}, no read of a unit vanishes, and $I + W^\ell_{\text{read}}[:, S]\, W^\ell_{\text{write}}[S, :]$ is invertible for every block ℓ and every set S of its units, since each determinant is a polynomial in the weights that equals 1 when the writes vanish. On an open set of inputs on which no unit of the residual network switches, with active sets S_ℓ, the network is the linear map $x \mapsto xEMD$ with $M = \prod_\ell \big(I + W^\ell_{\text{read}}[:, S_\ell]\, W^\ell_{\text{write}}[S_\ell, :]\big)$ invertible, of rank d_{model}. The inputs at which a unit switches lie in finitely many hyperplanes, by induction over the blocks: off the hyperplanes of blocks $1, \dots, \ell - 1$, the active sets of these blocks take finitely many values, and for each of them the argument of a unit of block ℓ is x times a vector $EM'w$, with M' invertible and $w \neq 0$ the read of the unit. So the inputs near which the network is linear of rank d_{model} form an open dense set Ω.

Let a skipless network F' have embedding E' and a first block with reads $w'_1, \dots, w'_{m'}$, $m' < 2d_{\text{model}}$, and set $a_i = E'w'_i$. Take x_0 with $x_0, -x_0 \in \Omega$ and $x_0 a_i \neq 0$ whenever $a_i \neq 0$. The units active near x_0, those with $x_0 a_i > 0$, and those active near $-x_0$ are disjoint, so one of the two sets, say the first after replacing x_0 by $-x_0$ if needed, has fewer than d_{model} units. Near x_0 the other units of the first block output zero, so F' depends on x only through the numbers xa_i of the active units, and $F'(x_0 + u) = F'(x_0)$ for every small u in a subspace of dimension greater than $d_{\text{in}} - d_{\text{model}}$. The residual network F changes by $uEMD$, which vanishes only on a subspace of dimension $d_{\text{in}} - d_{\text{model}}$. So some small u changes F and leaves F' unchanged, and $F' \neq F$.

GELU. As $\text{GELU}(y) - \text{ReLU}(y) = -|y|\, \Phi(-|y|)$, with Φ the standard normal distribution function, we have $|\text{GELU}(ty)/t - \text{ReLU}(y)| \leq \kappa/t$ for all $y \in \mathbb{R}$ and $t > 0$, where $\kappa = \max_{u\geq 0} u\, \Phi(-u) < 0.17$. Given a network of GELU blocks without normalization, let $h_\ell(t)$ be its hidden states at the input tx, and h^{R}_ℓ those of the ReLU network with the same weights at the input x. Then $h_0(t)/t = h^{\text{R}}_0$. If $h_{\ell-1}(t)/t = h^{\text{R}}_{\ell-1} + O(1/t)$, the output of block ℓ, divided by t, is $\text{GELU}\big(h_{\ell-1}(t)W^\ell_{\text{read}}\big)W^\ell_{\text{write}}/t = \text{ReLU}\big(h^{\text{R}}_{\ell-1}W^\ell_{\text{read}}\big)W^\ell_{\text{write}} + O(1/t)$, by the bound and since ReLU is 1-Lipschitz, so $h_\ell(t)/t = h^{\text{R}}_\ell + O(1/t)$, with or without the skip. So the output of the GELU network at tx, divided by t, tends to the output of the ReLU network at x. If a residual GELU network equals a skipless one, the two ReLU networks with the same weights are therefore equal. By the case of ReLU, the weights of the residual network for which this happens with a skipless network whose first block has fewer than $2d_{\text{model}}$ units form a set of measure zero. □

Proof of Proposition 4.3. Theorem 3.1(ii) with $M = I$ turns each residual block, whose read has rank d_{model}, into a skipless block with d_{model} more hidden units. Composing these blocks, with the same embedding and readout, gives a skipless network that equals the residual one. □

Proof of Proposition 4.4. As in the proof of Theorem 4.2, we compare the networks on rows $x \in \mathbb{R}^{1\times d_{\text{in}}}$, and GELU reduces to ReLU: if a residual GELU network equals a skipless one, the two ReLU

networks with the same weights are equal, with the same widths, and the condition on the norms does not involve the activation. So let σ be ReLU. Every block below is positively homogeneous, $B(ch) = cB(h)$ for $c > 0$, since there are no biases.

Step 1: the residual network is injective. Let $\Phi_\ell(h) = h + B_\ell(h)$. If $\Phi_\ell(h) = \Phi_\ell(h')$ with $h \neq h'$, then $\|h - h'\| = \|B_\ell(h') - B_\ell(h)\| \leq \|W^\ell_{\text{read}}\| \, \|W^\ell_{\text{write}}\| \, \|h - h'\| < \|h - h'\|$, as ReLU is 1-Lipschitz. So each Φ_ℓ is injective. As D has rank d_{model}, the residual network F satisfies $F(x) = F(x')$ only if $xE = x'E$.

Step 2: every skipless block is injective. Let a skipless network $F' = F$ have embedding E', blocks $B'_1, \ldots, B'_{L'}$ and readout D', and write $F'(x) = H(xE')$. If $xE' = x'E'$, then $F(x) = F(x')$, so $xE = x'E$: the kernel of E' lies in that of E. As E has rank d_{model}, so does E', the two kernels are equal, and H is injective on all of $\mathbb{R}^{1 \times d_{\text{model}}}$. In particular B'_1 is injective. A continuous, positively homogeneous, injective map ψ of $\mathbb{R}^{1 \times d_{\text{model}}}$ to itself is onto: $\psi(0) = 0$ by homogeneity, so $\psi(u) \neq 0$ for $u \neq 0$, so $\|\psi(u)\| \geq c\|u\|$ with $c > 0$ the minimum of $\|\psi\|$ on the unit sphere, hence the image of ψ is closed; by invariance of domain it is open; and it is not empty, so it is all of the connected space $\mathbb{R}^{1 \times d_{\text{model}}}$. So B'_1 is onto, B'_2 is injective on all of $\mathbb{R}^{1 \times d_{\text{model}}}$, and by induction every block B'_k is.

Step 3: an injective block has $2d_{\text{model}}$ units. Let B'_k have reads $r_1, \ldots, r_{m'}$, and drop the units with $r_i = 0$, which are never active. Fix u off the hyperplanes $\{u : ur_i = 0\}$; as they pass through 0, so is $-u$. Near each of the two points the block is linear, equal to $v \mapsto v \, W'_{\text{read}}[:, S] \, W'_{\text{write}}[S, :]$ with S the set of units active there. Injectivity makes this matrix invertible, so $|S| \geq d_{\text{model}}$ at u and at $-u$. The active sets at u and $-u$ are disjoint, so B'_k has at least $2d_{\text{model}}$ units. □

Remark A.12 (Folding). For ReLU, Proposition 4.4 fails without a condition on the blocks. Let E and D have rank d_{model}, let block 1 have d_{model} units with $W_{\text{read}} = -I$ and $W_{\text{write}} = 2I$, and let blocks $2, \ldots, L$ have d_{model} units each, with reads whose entries are all negative. Then $h + B_1(h) = (|h_1|, \ldots, |h_{d_{\text{model}}}|)$, so $(h + B_1(h))\, r < 0$ for every $h \neq 0$ and every read r of a later block. This is a maximum over the unit sphere, so it persists under small changes of all the weights, which form an open set; block 1 is not a contraction there, since $\|W_{\text{read}}\| \, \|W_{\text{write}}\|$ is near 2. On this set the units of later blocks are never active, and the residual network computes $x \mapsto \big(xE + B_1(xE)\big)D$. Let K be the closed convex cone spanned by the image of $h \mapsto h + B_1(h)$; every later read is negative on $K \setminus \{0\}$, so K is pointed. A skipless network equals the residual one with $2d_{\text{model}}$ units in its first block (Theorem 3.1(ii), as the read of block 1 has rank d_{model}) and d_{model} units in each later block: these compute the identity on K with linearly independent reads $v_1, \ldots, v_{d_{\text{model}}}$, positive on $K \setminus \{0\}$, and the writes $[\, v_1 \cdots v_{d_{\text{model}}} \,]^{-1}$. So the later blocks of this skipless network have fewer than $2d_{\text{model}}$ units.

A.2 No absorption in attention (Theorem 3.4, Proposition A.15)

Theorem 3.4 collects what the main text uses of the following theorem, for the attention blocks B_θ of (4).

Theorem A.13 (Attention and linear terms)**.** *Let θ and θ' be parameters of attention blocks with any numbers of heads and any head dimensions, and let $M \neq 0$ be a $d_{\text{model}} \times d_{\text{model}}$ matrix.*
(i) *For almost every $h \in \mathbb{R}^{T \times d_{\text{model}}}$, $hM + B_\theta(h) \neq B_{\theta'}(h)$.*
(ii) *Some h whose rows have norm at most 1 has a row in which $hM + B_\theta(h)$ and $B_{\theta'}(h)$ differ by at least $\|M\|/3$, where $\|M\|$ is the spectral norm.*
(iii) *With an embedding E and a readout D with $ED \neq 0$, the residual network $X \mapsto \big(XE + B_\theta(XE)\big)D$ differs from every skipless network $X \mapsto B_{\theta'}(XE')D'$.*

Overview. Two kinds of input suffice. If all tokens are equal, attention returns them unchanged, so the value–output products of all heads must add up to M. If one token is zero, all its scores are zero, so every head gives it the average of the tokens, while the skip gives it zero; this asks the same sum to vanish. Part (ii) makes the conflict quantitative, and analyticity turns it into part (i). With an embedding and a readout, the same two inputs give part (iii).

Proof of Theorem A.13. Let $F(h) = hM + B_\theta(h) - B_{\theta'}(h)$, and let θ' have H' heads. The difference $B_{\theta'} - B_\theta$ is one attention block with $H + H'$ heads,

$$B_{\theta'}(h) - B_\theta(h) = \sum_{k=1}^{H+H'} R_k(h)\, h\, C_k,$$

where the $R_k(h)$ are the attention maps of the heads of θ' and of θ, and the C_k are their value–output products, with a minus sign for the heads of θ. Let $C = \sum_k C_k$.

Step 1: equal tokens. Let every row of h be x. Each row of $R_k(h)$ sums to 1, so every row of $R_k(h)h$ is x. Hence every row of $F(h)$ is $x(M - C)$.

Step 2: one zero token. Let the last row of h be 0 and the other rows y. Every score in the last row of $R_k(h)$ is a product with the last row of h, which is 0. So that row of $R_k(h)$ is $(1/T, \dots, 1/T)$, also with a causal mask, since the last token sees all T tokens. Hence the last row of $F(h)$ is $-\frac{T-1}{T}\, yC$.

Proof of (ii). Over $\|x\|, \|y\| \le 1$, Step 1 gives rows of norm up to $\|M - C\|$, and Step 2 up to $\frac{T-1}{T}\|C\| \ge \frac{1}{2}\|C\|$. If $\|M - C\| < \|M\|/3$, then $\|C\| > \frac{2}{3}\|M\|$, so Step 2 reaches more than $\|M\|/3$.

Proof of (i). F is real-analytic, because the softmax is, and by (ii) some entry of F is not identically zero. The zeros of a real-analytic function that is not identically zero form a set of measure zero: in one variable they are isolated, and the general case follows by induction on the dimension with Fubini's theorem.

Proof of (iii). Let C_θ and $C_{\theta'}$ be the sums of the value–output products of the heads of θ and of θ'. On inputs whose rows all equal a row vector u, Step 1 gives the rows $uE(I + C_\theta)D$ in the residual network and $uE'C_{\theta'}D'$ in the skipless network. On inputs whose last row is 0 and whose other rows equal v, Step 2 gives the last rows $\frac{T-1}{T}\, vEC_\theta D$ and $\frac{T-1}{T}\, vE'C_{\theta'}D'$. If the two networks are equal, then $E(I + C_\theta)D = E'C_{\theta'}D' = EC_\theta D$, so $ED = 0$. □

We recall Theorem 3.4 from Section 3.

Theorem 3.4 (Attention blocks)**.** *Let θ and θ' be parameters of attention blocks with any numbers of heads and any head dimensions, and let $M \neq 0$ be a $d_{\text{model}} \times d_{\text{model}}$ matrix. For almost every $h \in \mathbb{R}^{T \times d_{\text{model}}}$, $hM + B_\theta(h) \neq B_{\theta'}(h)$.*

Proof of Theorem 3.4. It is Theorem A.13(i). □

Proposition A.15 follows from a lemma on attention heads.

Lemma A.14 (Attention heads are identifiable)**.** *Let $T \ge 3$. Let $\Gamma_1, \dots, \Gamma_m \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ be distinct, let $C_1, \dots, C_m, M \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$, and write $P_\Gamma(h) = \operatorname{softmax}(h\Gamma h^\top)$, with the softmax taken row by row. If*

$$hM + \sum_{k=1}^{m} P_{\Gamma_k}(h)\, h\, C_k = 0$$

for all h in a nonempty open subset of $\mathbb{R}^{T \times d_{\text{model}}}$, then $M = 0$ and $C_1 = \cdots = C_m = 0$.

Overview. Move one token along a complex line. Each head becomes a meromorphic function of one complex variable, whose poles lie on a vertical line that its score matrix determines, so heads with distinct score matrices have disjoint poles. At a pole of one head the other terms are regular, so its residue must vanish, and the residues force its value–output product to vanish. What is left is the skip and a uniform head, which two simple inputs separate.

Proof of Lemma A.14. One moving token. Fix h in the open set, with rows $x_1, \dots, x_T$, and fix $v \in \mathbb{R}^{1 \times d_{\text{model}}}$. Replace the second row by $x_2 + tv$ with $t \in \mathbb{C}$, and look at the first row. For a head with score matrix Γ, the first row has scores $z_r = x_1 \Gamma x_r^\top$ for $r \neq 2$, and $a + bt$ for $r = 2$, where $a = x_1 \Gamma x_2^\top$ and $b = x_1 \Gamma v^\top$. Its output is

$$g_\Gamma(t) = \frac{e^{a+bt}\,(x_2 + tv) + Z\bar{x}}{e^{a+bt} + Z}, \qquad Z = \sum_{r \neq 2} e^{z_r}, \qquad \bar{x} = \frac{1}{Z} \sum_{r \neq 2} e^{z_r} x_r,$$

where $\bar{x}$ is the average of the other tokens under the weights of the head. The first row of hM is $x_1 M$, which does not depend on t.

Poles. If $b \neq 0$, then g_Γ is meromorphic on $\mathbb{C}$, with simple poles at

$$t_n = c + \frac{(2n+1)\pi i}{b}, \quad n \in \mathbb{Z}, \qquad \text{where } c = \frac{\log Z - a}{b},$$

so all its poles lie on the vertical line $\operatorname{Re} t = c$. The residue at t_n is $(x_2 + t_n v - \bar{x})/b$. If $\Gamma = 0$, then g_Γ is affine in t and has no poles.

The poles tell the heads apart. For h and v outside a closed set with empty interior, $b \neq 0$ for every $\Gamma_k \neq 0$, and distinct $\Gamma_k \neq 0$ give distinct lines $\operatorname{Re} t = c$. The first condition fails only on the zero sets of nonzero polynomials. For the second, suppose that two heads $\Gamma \neq \Gamma'$ have $c = c'$ for all h and v near some point. Clearing denominators and comparing the terms bilinear in (x_2, v) gives $x_1\Gamma' = \lambda\, x_1\Gamma$ for some number $\lambda \neq 0$. Comparing the terms linear in v then gives $\log Z' = \lambda \log Z$, that is,

$$\log \sum_{r \neq 2} e^{\lambda z_r} \;=\; \lambda \log \sum_{r \neq 2} e^{z_r}.$$

Because $T \geq 3$, the sum has at least two terms, $r = 1$ and $r = 3$, and z_3 varies with x_3. Differentiating in z_3 gives $\operatorname{softmax}(\lambda z)_3 = \operatorname{softmax}(z)_3$ on an interval of z_3, which forces $\lambda = 1$. So $x_1\Gamma' = x_1\Gamma$ for all x_1 near a point, and $\Gamma' = \Gamma$, a contradiction. With two tokens the sum has the single term $r = 1$, and this step fails (Section 3).

Residues. The identity holds for real t near 0, so by analytic continuation it holds on all of $\mathbb{C}$ away from the poles. At a pole t_n of a head $\Gamma_k \neq 0$, no other term is singular, so the residue times C_k vanishes: $(x_2 + t_n v - \bar{x})\, C_k = 0$ for every $n \in \mathbb{Z}$. Subtracting two consecutive ones gives $vC_k = 0$. This holds for all v in an open set, so $C_k = 0$.

The skip. What remains is $hM + P_0(h)\, h\, C_0 = 0$, where C_0 is the coefficient of $\Gamma = 0$ (or 0 if no Γ_k vanishes). This is linear in h, so it holds for all h. Take the first row of h equal to x and the other rows 0: the first two rows give $x(M + C_0/T) = 0$ and $xC_0/T = 0$. So $C_0 = 0$ and $M = 0$. □

The lemma gives the proposition that Section 3.3 cites.

Proposition A.15 (The softmaxes must match)**.** *Let $T \geq 3$, let θ and θ' be parameters of attention blocks, and suppose that $hM + B_\theta(h) = B_{\theta'}(h)$ for all h in a nonempty open set. Then $M = 0$, and for every matrix Γ, the heads with score matrix Γ have the same sum of value–output products in B_θ as in $B_{\theta'}$.*

Proof of Proposition A.15. Group the heads of $B_{\theta'}$ and B_θ by score matrix, add their value–output products, with a minus sign for the heads of θ, and apply Lemma A.14. □

With two tokens, heads can cancel: the softmax of two scores is the logistic function of their difference, and $1/(1+e^{-z}) + 1/(1+e^{z}) = 1$ makes two heads with opposite score matrices add up to a uniform head. A skip still cannot be absorbed when $T = 2$ (Theorem 3.4).

A.3 Networks under RMSNorm (Theorem 4.5)

We recall Theorem 4.5 from Section 4.

Theorem 4.5 (Networks under RMSNorm)**.** *Consider pre-RMSNorm networks, with $\varepsilon \geq 0$ and blocks of any kind and width. For almost every choice of weights, a residual network equals no skipless network, whatever its depth, widths and kinds of blocks.*

Write N_0 for RMSNorm with $\varepsilon = 0$, which maps each nonzero row w to $\sqrt{d_{\text{model}}}\, w/\|w\|$, with the derivative $\mathrm{D}N_0(w)[b] = \sqrt{d_{\text{model}}}\,\big(b - (b \cdot \hat{w})\,\hat{w}\big)/\|w\|$, where $\hat{w} = w/\|w\|$; it vanishes exactly when b is a multiple of w. Outside a set of weights of measure zero, E and D have rank d_{model}, so that XE takes every value in $\mathbb{R}^{T \times d_{\text{model}}}$ and $vD = 0$ only for $v = 0$. Each block is linear in its write, which for an attention block holds the matrices $W_{O,i}$: block ℓ adds $a_\ell W^\ell_{\text{write}}$ to the hidden state, where a_ℓ, a function of the normalized state n that block ℓ reads, holds the activations of its hidden

units, or for an attention block the matrices $P_{\Gamma_i}(n)\, n\, W_{V,i}$ of its heads side by side. It depends on the read of block ℓ only.

Proof of Theorem 4.5. Fix an input X for which XE has no zero row, and let $t \to \infty$.

The skipless network. Its hidden states depend on the input only through $N(tXE')$, at its embedding E'. On a row w of XE', $N(tw) = w/\sqrt{\|w\|^2/d_{\text{model}} + \varepsilon/t^2}$. For $\varepsilon = 0$ this is $N(w)$, so the output at tX is the output at X. For $\varepsilon > 0$ it is a smooth function of ε/t^2 that equals $N_0(w)$ at 0, or 0 if $w = 0$, and the rest of the network is locally Lipschitz, so the output at tX is $G(X) + O(1/t^2)$ for some $G(X)$.

The residual network. Let $S = N_0(XE)$ and $\beta(X) = \sum_\ell a_\ell(S) W^\ell_{\text{write}}$, the sum of the outputs of all blocks at the normalized state S. By induction over the blocks, $h_\ell = tXE + \sum_{k\le\ell} a_k(S) W^k_{\text{write}} + O(1/t)$: each hidden state is $tXE + O(1)$, so for large t it has no zero row, the next block reads $N(h_\ell) = S + O(1/t)$, and, being locally Lipschitz, it adds its output at S up to $O(1/t)$. At the readout, on each row, $N(tw + b + O(1/t)) = N_0\big(w + b/t + O(1/t^2)\big) + O(1/t^2) = N_0(w) + \mathrm{D}N_0(w)[b]/t + O(1/t^2)$. So the output at tX is

$$N_0(XE)D + \rho(X)/t + O(1/t^2), \qquad \rho(X) = \mathrm{D}N_0(XE)\big[\beta(X)\big]\, D,$$

with $\mathrm{D}N_0$ taken row by row.

Comparison. Let the two networks be equal. Both are continuous on the open set of inputs at which both are defined, so they agree on all of it. The skipless network is defined at tX if and only if it is defined at X, by the case $\varepsilon = 0$ above, and everywhere if $\varepsilon > 0$; the residual network is defined at tX for all large t. So if the skipless network is defined at X, their outputs at tX agree for all large t, and $\rho(X) = 0$. These inputs X are dense, and ρ is continuous on the inputs for which XE has no zero row, so $\rho = 0$ there. As D has rank d_{model}, each row of $\beta(X)$ is then a multiple of the same row of XE. Here $\beta(X)$ depends on X only through S, and as XE takes every value without zero rows, S takes every value whose rows have norm $\sqrt{d_{\text{model}}}$. So equality requires that for every such S, each row of $\sum_\ell a_\ell(S) W^\ell_{\text{write}}$ is a multiple of the same row of S.

Measure zero. Fix all weights but the writes. The condition is linear in the writes, so the writes that satisfy it form a linear subspace. The subspace is proper if some $a_\ell(S)$ has a nonzero entry (r, j) at some S: the write of block ℓ whose row j is not a multiple of row r of S, with all other entries and all other writes zero, violates the condition. And a_1 vanishes for all S only if the read of block 1 has a zero column, which happens on a set of measure zero. The weights that satisfy the condition form a closed set, since every a_ℓ is continuous in the read and in S, and by Fubini's theorem this set has measure zero. □

A.4 Skipless networks as limits of residual networks (Proposition 5.1)

Proposition 5.1 (Skipless networks as limits of residual networks)**.** *Let F be a skipless network whose blocks are locally Lipschitz.*

(i) Without normalization. *The network F is a limit of residual networks with blocks of the same kinds and widths, uniformly on bounded sets of inputs.*

(ii) Under pre-RMSNorm with $\varepsilon = 0$. *The network F is a limit of residual networks with blocks of the same kinds and widths, uniformly on compact sets of inputs at which F is defined.*

Let F have the embedding E, the blocks B_ℓ, with reads W^ℓ_{read} and writes W^ℓ_{write}, and the readout D, and let $s_0 = XE$ and $s_\ell = B_\ell(s_{\ell-1})$ be its hidden states, so that $F(X) = s_L D$. For $t \ge 1$, let F_t be the residual network with the embedding E, the reads $t^{1-\ell} W^\ell_{\text{read}}$, the writes $t^\ell W^\ell_{\text{write}}$ and the readout $t^{-L} D$; under pre-RMSNorm with $\varepsilon = 0$, let F_t keep the embedding, the reads and the readout of F and have the writes $t^\ell W^\ell_{\text{write}}$. Let $r_0 = 0$ and

$$r_\ell \;=\; B_\ell(s_{\ell-1} + r_{\ell-1}) - B_\ell(s_{\ell-1}) \;+\; t^{-1}\,(s_{\ell-1} + r_{\ell-1}). \tag{9}$$

The proofs show that the hidden states of F_t are $t^\ell(s_\ell + r_\ell)$ and bound r_ℓ, which shows that F_t tends to F. Norms are Frobenius norms for hidden states and operator norms for matrices.

Proof of Proposition 5.1(i). Block ℓ of F_t has its reads scaled by $t^{1-\ell}$ and its writes by t^ℓ, so it maps h to $t^\ell B_\ell(t^{1-\ell}h)$. We show by induction that its hidden states are $h_\ell = t^\ell(s_\ell + r_\ell)$, with $r_0 = 0$ and r_ℓ given by (9). For $\ell = 0$, $h_0 = XE = s_0$. If $h_{\ell-1} = t^{\ell-1}(s_{\ell-1} + r_{\ell-1})$, then

$$h_\ell \;=\; h_{\ell-1} + t^\ell B_\ell(t^{1-\ell}h_{\ell-1}) \;=\; t^\ell\big(B_\ell(s_{\ell-1}+r_{\ell-1}) + t^{-1}(s_{\ell-1}+r_{\ell-1})\big) \;=\; t^\ell(s_\ell + r_\ell).$$

Let $\mathcal{X}$ be a bounded set of inputs and $\sigma_\ell = \sup_{X\in\mathcal{X}}\|s_\ell\|$, which is finite because the blocks are continuous. Suppose that $\|r_{\ell-1}\| \le C_{\ell-1}/t$ for all $X \in \mathcal{X}$ and $t \ge 1$, which holds for $\ell = 1$ with $C_0 = 0$. Then $s_{\ell-1}$ and $s_{\ell-1} + r_{\ell-1}$ lie in the ball of radius $\sigma_{\ell-1} + C_{\ell-1}$, on which B_ℓ is Lipschitz with some constant Λ_ℓ, since B_ℓ is locally Lipschitz. Hence

$$\|r_\ell\| \;\le\; \Lambda_\ell\|r_{\ell-1}\| + t^{-1}\big(\sigma_{\ell-1} + \|r_{\ell-1}\|\big) \;\le\; C_\ell/t, \qquad C_\ell = \Lambda_\ell C_{\ell-1} + \sigma_{\ell-1} + C_{\ell-1}.$$

Finally $F_t(X) = t^{-L}h_L D = F(X) + r_L D$, so $\|F_t(X) - F(X)\| \le C_L\|D\|/t$ for all $X \in \mathcal{X}$ and $t \ge 1$. □

Proof of Proposition 5.1(ii). Let $\mathcal{X}$ be a compact set of inputs at which F is defined, that is, at which no row of XE or of a hidden state of F vanishes. Here N divides each row of h by its norm and multiplies it by $\sqrt{d_{\text{model}}}$, so $N(ch) = N(h)$ for $c > 0$, and $\|N(y') - N(y)\| \le 2\sqrt{d_{\text{model}}}\,\|y' - y\|/\|y\|$ for nonzero rows y, y'. The hidden states of F are $s_0 = XE$ and $s_\ell = B_\ell(N(s_{\ell-1}))$, and $F(X) = N(s_L)D$. As no row of s_ℓ vanishes on $\mathcal{X}$, continuity and compactness give $\delta > 0$ below the norm of every row of every s_ℓ, $\ell \ge 1$, on $\mathcal{X}$. Since N removes the factor $t^{\ell-1}$, the hidden states of F_t are again $h_0 = s_0$ and $h_\ell = t^\ell(s_\ell + r_\ell)$, now with $r_0 = 0$ and

$$r_\ell \;=\; B_\ell\big(N(s_{\ell-1}+r_{\ell-1})\big) - B_\ell\big(N(s_{\ell-1})\big) \;+\; t^{-1}\,(s_{\ell-1}+r_{\ell-1}),$$

as long as no row of $s_{\ell-1} + r_{\ell-1}$ vanishes. The maps B_ℓ are Lipschitz on the bounded image of N, with some constants Λ_ℓ, and $\sigma_\ell = \sup_{X\in\mathcal{X}}\|s_\ell\|$ is finite. If every row of $r_{\ell-1}$ is shorter than δ, then $\|r_\ell\| \le 2\sqrt{d_{\text{model}}}\,\Lambda_\ell\|r_{\ell-1}\|/\delta + t^{-1}(\sigma_{\ell-1} + \|r_{\ell-1}\|)$, so by induction $\|r_\ell\| \le C_\ell/t$ on $\mathcal{X}$ for $t \ge 1$, with $C_0 = 0$ and $C_\ell = (2\sqrt{d_{\text{model}}}\,\Lambda_\ell/\delta + 1)C_{\ell-1} + \sigma_{\ell-1}$, and every row of r_ℓ is shorter than δ once $t \ge t_0 = \max(1, \max_\ell 2C_\ell/\delta)$. Finally $\|F_t(X) - F(X)\| = \|(N(s_L + r_L) - N(s_L))D\| \le 2\sqrt{d_{\text{model}}}\,\|D\|\,C_L/(\delta t)$ for all $X \in \mathcal{X}$ and $t \ge t_0$. □

Remark A.16 (The rate). The proofs give the rate $1/t$. Without normalization, for every bounded set $\mathcal{X}$ of inputs there is C such that $\|F_t(X) - F(X)\| \le C/t$ for all $X \in \mathcal{X}$ and $t \ge 1$. Under pre-RMSNorm with $\varepsilon = 0$, the same holds for all $t \ge t_0$, for every compact set $\mathcal{X}$ of inputs at which F is defined, with t_0 depending on $\mathcal{X}$. The constants depend on $\mathcal{X}$ and on F.

B Universal approximation: width and biases

Lin and Jegelka (2018) show that residual ReLU networks whose blocks have a single hidden unit, $h \mapsto h + \mathrm{ReLU}(hw + c)\,v$ with $w \in \mathbb{R}^{d\times 1}$, $v \in \mathbb{R}^{1\times d}$ and a bias $c \in \mathbb{R}$, approximate every integrable function on $\mathbb{R}^d$ in L^1 as the depth grows, while ReLU networks without skips whose hidden layers have width at most the input dimension d cannot (their Proposition 2.1). The separation is confined to this narrow regime: with hidden layers of width $d + 1$, ReLU networks without skips already approximate every continuous function uniformly on compact sets (Hanin and Sellke, 2017).

The construction also needs the biases, for the shifts and the cut-offs at constants of its blocks (their Proposition 3.2). Without biases, a ReLU network, with or without skips, is positively homogeneous of degree one, $F(\lambda x) = \lambda F(x)$ for $\lambda > 0$, since ReLU and the linear maps are. On every ray $\{t\theta : t > 0\}$ such a network is a multiple of t, and so is every limit of such networks, uniformly or in L^1 on a compact set, on almost every ray: the multiples of t form a closed space, and by Fubini's theorem L^1 convergence passes to almost every ray. In one dimension, for instance, every such network is $x \mapsto a\,\mathrm{ReLU}(x) - b\,\mathrm{ReLU}(-x)$, which stays at L^1 distance at least $(2 - \sqrt{2})/3$ from x^2 on $[-1, 1]$. On all of $\mathbb{R}^d$, no integrable function other than 0 can be approximated in L^1, since a positively homogeneous function other than 0 is not integrable.

C Blocks with biases

The main text treats blocks without biases, as in many current language models. With biases the answers change: every ReLU, GELU, ReLU^2 and gated block absorbs a linear term with more units, while $\tanh$ and attention blocks still never do (Theorem C.1). To absorb its skip, almost every ReLU, GELU and gated block needs exactly d_{model} more units, and almost every ReLU^2 block between $2d_{\text{model}}$ and $3d_{\text{model}}$ (Theorem C.2). Table 2 collects these results.

Blocks with biases. A vanilla MLP block with biases computes $B_\theta(x) = f(xW_{\text{read}} + b)\, W_{\text{write}} + c$, with $\theta = (W_{\text{read}}, b, W_{\text{write}}, c)$, a unit bias $b \in \mathbb{R}^{1\times m}$ and an output bias $c \in \mathbb{R}^{1\times d_{\text{model}}}$; unit i has the read w_i, the bias b_i and the write $\hat{w}_i$. In a gated block with biases, unit j has a gate read g_j (a column of W_{g}) and a gate bias b^{g}_j, a value read v_j (a column of W_{v}) and a value bias b^{v}_j, and a write $\hat{w}_j$, and

$$B_\theta(x) = \sum_{j=1}^{m} \sigma\left(xg_j + b^{\text{g}}_j\right)\left(xv_j + b^{\text{v}}_j\right)\hat{w}_j + c.$$

An attention block with biases adds a bias row to every query, key and value and to its output: with $\mathbf{1}$ the column of T ones,

$$B_\theta(h) = \sum_{i=1}^{H} P_i(h)\left(hW_{V,i} + \mathbf{1}b_{V,i}\right) W_{O,i} + \mathbf{1}b_O,$$

where $P_i(h)$ is the softmax, row by row, of $(hW_{Q,i}+\mathbf{1}b_{Q,i})(hW_{K,i}+\mathbf{1}b_{K,i})^\top/\sqrt{d_h}$, with or without a causal mask. These scores do not depend on the positions of the tokens, which excludes rotary embeddings. A block absorbs xM, for $M \neq 0$, if a block of the same kind with biases satisfies (2), or for attention, with the same mask, $hM + B_\theta(h) = B_{\theta'}(h)$ for all $h \in \mathbb{R}^{T\times d_{\text{model}}}$.

Theorem C.1 (Exact absorption with biases)**.** *Let blocks have biases, and let $M \neq 0$.*
(i) *Every ReLU, GELU, ReLU^2, ReGLU, SwiGLU or GeGLU block absorbs xM with more hidden units.*
(ii) *No $\tanh$ block and no attention block absorbs xM, at any width.*
(iii) *For generic θ, no block $B_{\theta'}$ of the same width as B_θ satisfies (2).*

Part (iii) follows from (ii) for $\tanh$ and attention blocks, and from the proofs of Theorem C.2 for the others.

Theorem C.2 (Extra units with biases)**.** *Let blocks have biases, let M be invertible and let θ be generic.*
(i) *A ReGLU, SwiGLU or GeGLU block absorbs xM with d_{model} more hidden units, and not with fewer.*
(ii) *A ReLU or GELU block with at least d_{model} hidden units absorbs xM with d_{model} more hidden units, and not with fewer.*
(iii) *A ReLU^2 block with at least d_{model} hidden units absorbs xM with $3d_{\text{model}}$ more hidden units, and not with fewer than $2d_{\text{model}}$.*

Remark C.3 (Constructions for every block). Every gated block absorbs xM with $\operatorname{rank} M$ more units, and for generic θ fewer do not (proof of Theorem C.2(i)). Every ReLU or GELU block absorbs xM with $2d_{\text{model}}$ more units, and with d_{model} when its read has rank d_{model}. Every ReLU^2 block absorbs xM with $4d_{\text{model}}$ more units, and with $3d_{\text{model}}$ when its read has rank d_{model}. Almost every ReLU^2 block with at least $d_{\text{model}}(d_{\text{model}}+3)/2$ units absorbs it with $d_{\text{model}}(d_{\text{model}}+3)/2$ more, which is fewer than $3d_{\text{model}}$ only for $d_{\text{model}} = 2$, where it is five.

Remark C.4 (Bounded sets of inputs). GELU, SwiGLU, GeGLU, $\tanh$ and attention blocks are real-analytic in their input, so an equality on a nonempty open set of inputs holds everywhere, and Theorems C.1 and C.2 hold on every such set. ReLU, ReLU^2 and ReGLU blocks are not. On a ball $\mathcal{X}$ of inputs, a unit whose argument (for ReGLU, whose gate's argument) does not vanish

Proven	**ReLU**	**GELU**	**tanh**	**ReLU^2**	**ReGLU, SwiGLU, GeGLU**	**Attention**
one block, same width	measure zero Thm. C.1(iii)		never Thm. C.1(ii)	measure zero Thm. C.1(iii)		never Thm. C.1(ii)
one block, more hidden units or heads	exactly d_{model} more Thm. C.2(ii)		never Thm. C.1(ii)	$2d_{\text{model}}$ to $3d_{\text{model}}$ more Thm. C.2(iii)	exactly d_{model} more Thm. C.2(i)	never Thm. C.1(ii)

Table 2: What is proven for one block with biases. Never: for no weights. Measure zero: only for a null set of weights. Exactly d_{model} more: for almost every block, d_{model} more hidden units suffice and fewer do not. Scope in Section 2.

on $\mathcal{X}$ is a polynomial there. If d_{model} units of a ReLU block with linearly independent reads, or $d_{\text{model}}(d_{\text{model}}+3)/2$ generic units of a ReLU^2 or ReGLU block, are active on all of $\mathcal{X}$, changing their writes and the output bias absorbs xM on $\mathcal{X}$ at the same width. So on $\mathcal{X}$, the entries "measure zero" of Table 2 fail for ReLU, ReLU^2 and ReGLU blocks, for an open set of blocks. The lower bounds below use inputs of unbounded size: rays $x = tu$ with $t \to \infty$, or the hyperplanes on which the arguments of the units vanish, wherever they lie.

Remark C.5 (Networks). Without normalization, absorbing the skip block by block, with $M = I$, shows that every residual network of ReLU^2, ReGLU, SwiGLU or GeGLU blocks with biases equals a skipless network with $4d_{\text{model}}$ (ReLU^2) or d_{model} (gated) more units per block; without biases, none with $ED \neq 0$ does (Theorem 4.1(i)). As without biases, every residual ReLU or GELU network whose reads have rank d_{model} equals a skipless one with d_{model} more units per block. A skipless network of $\tanh$ blocks with biases is bounded, so it equals no residual one with $ED \neq 0$. Networks with biases under pre-RMSNorm are not treated here.

Open questions. The exact number of extra units for ReLU^2 blocks lies between $2d_{\text{model}}$ and $3d_{\text{model}}$; for $d_{\text{model}} = 2$, five suffice once the block has five units (Remark C.3). Attention with biases and rotary embeddings is open.

C.1 Constructions and blocks that never absorb

Let e_a be the a-th standard basis column of $\mathbb{R}^{d_{\text{model}}}$ and M_a the a-th row of M; then xM is the sum over a of $(xe_a)\, M_a$.

Proof of Theorem C.1(i). The constructions also give the upper bounds of Theorem C.2. Each construction keeps the units of B_θ, changes some of their writes and the output bias, and adds units.

Gated. A unit with gate read 0 and gate bias 1 computes $\sigma(1)\,(xv + b^{\text{v}})\,\hat{w}$, which is affine in x, and $\sigma(1) \neq 0$ for ReLU, SiLU and GELU. Write M as a sum of $\operatorname{rank} M$ matrices of rank one, each a column times a row. For each term, a unit with gate read 0, gate bias 1, value bias 0, the column as its value read and the row divided by $\sigma(1)$ as its write adds x times the term. So $\operatorname{rank} M \leq d_{\text{model}}$ units add xM.

ReLU and GELU. Both satisfy (3). Two new units with reads $\pm e_a$, biases 0 and writes $\pm M_a$ add $(xe_a)\, M_a$, so $2d_{\text{model}}$ new units add xM. If the read has rank d_{model}, choose d_{model} units with linearly independent reads, and let the rows γ_i, for these units i, solve $\sum_i w_i\gamma_i = M$. Give unit i the write $\hat{w}_i + \gamma_i$, and add a unit with read $-w_i$, bias $-b_i$ and write $-\gamma_i$. By (3) the two add $(xw_i + b_i)\,\gamma_i$, and subtracting $\sum_i b_i\gamma_i$ from the output bias leaves xM, with d_{model} new units.

ReLU^2. Let $r(z) = \max(z, 0)^2$, so that $r(z) + r(-z) = z^2$. Four new units compute $r\big(\pm(xe_a + 1)\big)$, with the write $M_a/4$, and $r\big(\pm(xe_a - 1)\big)$, with the write $-M_a/4$. Together they add $\big((xe_a+1)^2 - (xe_a-1)^2\big)M_a/4 = (xe_a)\, M_a$, so $4d_{\text{model}}$ new units add xM. If the read has rank d_{model}, take the units i and the rows γ_i as above, and let $z_i = xw_i + b_i$. Give unit i the write $\hat{w}_i + \gamma_i$ and add three units: one with read $-w_i$, bias $-b_i$ and write γ_i, and two with reads $\pm w_i$, biases $\pm(b_i - \frac{1}{2})$ and writes $-\gamma_i$. Together they add $\gamma_i\big(z_i^2 - (z_i - \frac{1}{2})^2\big) = (xw_i)\,\gamma_i + (b_i - \frac{1}{4})\,\gamma_i$, and the output bias absorbs the constants: $3d_{\text{model}}$ new units. Finally, for $d_{\text{model}}(d_{\text{model}}+3)/2$ units i of the block and new rows γ_i, give unit i the write $\hat{w}_i + \gamma_i$ and add a unit with read $-w_i$, bias $-b_i$ and write γ_i: these add $\sum_i \gamma_i z_i^2$. For each output coordinate o, $\sum_i \gamma_i z_i^2 = xM + \text{const}$ is a linear system of

$d_{\text{model}}(d_{\text{model}}+3)/2$ equations, one for each coefficient of a polynomial of degree two other than the constant, in the $d_{\text{model}}(d_{\text{model}}+3)/2$ unknowns $\gamma_{i,o}$. For generic θ it has a solution. □

Proof of Theorem C.1(ii). tanh. $B_{\theta'} - B_\theta$ is bounded, and xM is not.

Attention. Suppose that $hM + B_\theta(h) = B_{\theta'}(h)$ for all h, and let C and C' be the sums of the value–output products $W_{V,i}W_{O,i}$ of the heads of θ and of θ'. If all rows of h equal x, all keys of a head are equal, so each row of $P_i(h)$ is uniform on the tokens it sees, and all values equal $xW_{V,i} + b_{V,i}$. So every row of $B_\theta(h)$ is xC plus a constant row, and the equation gives $M = C' - C$. Now add $\epsilon\delta$ to the first row of this h, for a row δ, and look at the second row, which sees the first token with or without a causal mask. The second row of hM does not change. In the second row of $B_\theta(h)$, the derivatives of the attention weights at $\epsilon = 0$ sum to zero over the tokens and multiply equal values, so the derivative in ϵ at $\epsilon = 0$ is $\delta C/T_2$, with T_2 the number of tokens that the second token sees. Hence $\delta C = \delta C'$ for every δ, so $C = C'$ and $M = 0$, a contradiction. □

C.2 Lower bounds

For a unit j of a gated block with biases, let $\Delta_j = \left(b^{\text{g}}_j v_j + b^{\text{v}}_j g_j\right) \hat{w}_j$, a $d_{\text{model}} \times d_{\text{model}}$ matrix of rank at most one.

Lemma C.6 (Gated blocks with biases)**.** *Let ReGLU, SwiGLU or GeGLU blocks B_θ and $B_{\theta'}$ with biases satisfy (2), where $B_{\theta'}$ has n more units than B_θ. Let the gate reads of B_θ be nonzero and pairwise non-collinear, and its value reads and writes nonzero. Then for some set S of units of B_θ, the matrix $M + \sum_{i\in S} \Delta_i$ is a sum of at most n matrices of rank one.*

Proof. Outline. Along rays $x = tu$ with $t \to \infty$, every gate becomes a ReLU gate. The terms of order t^2 and t of (2) then split hyperplane by hyperplane, and those of order t express M through the units on one side of each hyperplane.

Expansion along rays. Group the units of both blocks with a nonzero gate read g by the hyperplane $\mathcal{H} = \{x : xg = 0\}$, choose a unit normal ν of $\mathcal{H}$, and write $g = s\eta\nu$ with $\eta > 0$ and a side $s = \pm 1$. Let $[\,\cdot\,]$ be 1 if the condition inside holds and 0 otherwise. For $x \notin \mathcal{H}$ and $t \to \infty$,

$$\sigma\left(txg + b^{\text{g}}\right) = t\eta\,\text{ReLU}(s\,x\nu) + b^{\text{g}}\,[s\,x\nu > 0] + o(1/t),$$

exactly for large t when σ is ReLU, and up to terms that decay exponentially in t when σ is SiLU or GELU. So the unit contributes

$$t^2\,\text{ReLU}(s\,x\nu)\,(xv)\,\eta\hat{w} + t\left(\text{ReLU}(s\,x\nu)\,\eta b^{\text{v}}\hat{w} + [s\,x\nu > 0]\,(xv)\,b^{\text{g}}\hat{w}\right) + O(1),$$

and a unit with gate read 0 contributes $t\,\sigma(b^{\text{g}})\,(xv)\,\hat{w} + O(1)$. Put tx into (2) and compare the coefficients of t^2 and of t at each x off all the hyperplanes. For a hyperplane $\mathcal{H}$ and a side $\pm$, let $A^\pm$, $a^\pm$ and $G^\pm$ be the sums of $\eta v\hat{w}$, $\eta b^{\text{v}}\hat{w}$ and $b^{\text{g}}v\hat{w}$ over the units of $B_{\theta'}$ on that side, minus the same sums over the units of B_θ, and let Z be the same difference for the sums of $\sigma(b^{\text{g}})\,v\hat{w}$ over the units with gate read 0. Then, with sums over the hyperplanes,

$$\begin{aligned} 0 &= \textstyle\sum_{\mathcal{H}} \left(\text{ReLU}(x\nu)\,xA^+ + \text{ReLU}(-x\nu)\,xA^-\right), \\ xM &= \textstyle\sum_{\mathcal{H}} \left(\text{ReLU}(x\nu)\,a^+ + \text{ReLU}(-x\nu)\,a^- + [x\nu > 0]\,xG^+ + [x\nu < 0]\,xG^-\right) + xZ. \end{aligned}$$

One hyperplane at a time. Near a point of $\mathcal{H}$ that lies on no other hyperplane, the terms of the other hyperplanes are smooth. In the second identity, continuity across $\mathcal{H}$ forces $yG^+ = yG^-$ for every row y with $y\nu = 0$, so $G^+ - G^- = \nu q$ for a row q. Then $[x\nu > 0]\,x\nu q = \text{ReLU}(x\nu)\,q$, and since the identity has no kink at $\mathcal{H}$, $a^+ + q + a^- = 0$. The terms of $\mathcal{H}$ are then $x\Delta_{\mathcal{H}}$, with

$$\Delta_{\mathcal{H}} = G^- - \nu a^- = G^+ + \nu a^+,$$

and the second identity becomes $M = \sum_{\mathcal{H}} \Delta_{\mathcal{H}} + Z$. As $g = \pm\eta\nu$ on side $\pm$, $G^\pm \pm \nu a^\pm$ is the sum of the matrices Δ_j over the units of $B_{\theta'}$ on side $\pm$, minus the same sum over the units of B_θ. In the

first identity, the same argument applied to $\mathrm{ReLU}(\pm x\nu) = (|x\nu| \pm x\nu)/2$ gives $A^+ + A^- = \nu p$ for a row p, and then $p = 0$, since $|x\nu|\, x\nu$ has a jump in its second derivative. So $A^+ + A^- = 0$.

Counting. Orient the hyperplane $\mathcal{H}_i$ of unit i of B_θ so that unit i lies on side $+$; no other unit of B_θ lies on $\mathcal{H}_i$. Let $J^\pm$ be the sets of units of $B_{\theta'}$ on side $\pm$ of $\mathcal{H}_i$. If both were empty, $A^+ + A^-$ would be $-\eta_i v_i \hat{w}_i$, where $g_i = \eta_i \nu$, which does not vanish; so $\mathcal{H}_i$ carries $|J^+| + |J^-| - 1 \geq 0$ extra units. If J^+ is empty, $\Delta_{\mathcal{H}_i} = -\Delta_i$. Otherwise $\Delta_{\mathcal{H}_i}$ is the sum of the Δ_j over J^-, at most $|J^-| \leq |J^+| + |J^-| - 1$ matrices of rank one. A hyperplane of no unit of B_θ carries $|J^+| + |J^-|$ units of $B_{\theta'}$, with $J^\pm$ defined in the same way, and $\Delta_{\mathcal{H}}$ is a sum of at most that many matrices of rank one. Each unit of $B_{\theta'}$ with gate read 0 adds one matrix of rank one to Z, and B_θ has no such unit. With S the units i for which J^+ is empty, $M + \sum_{i \in S} \Delta_i$ is therefore a sum of at most n matrices of rank one. □

Proof of Theorem C.2(i). The argument also gives Theorem C.1(iii) for gated blocks. The construction above gives d_{model} units. For each set S of units of B_θ, the determinant of $M + \sum_{i \in S} \Delta_i$ is a polynomial in θ, and it equals $\det M \neq 0$ when all biases vanish. So for generic θ it is nonzero for each of the finitely many sets S, the hypotheses of Lemma C.6 hold, and the lemma gives $n \geq d_{\mathrm{model}}$. For $M \neq 0$ of any rank, a minor of size $\operatorname{rank} M$ that does not vanish at M plays the role of the determinant, and gives $n \geq \operatorname{rank} M \geq 1$. □

Proof of Theorem C.2(ii). The argument also gives Theorem C.1(iii) for ReLU and GELU blocks. The construction above gives d_{model} units. For the converse, let $B_{\theta'}$ satisfy (2). ReLU changes by at most $|b|$ when its argument changes by b, and GELU differs from ReLU by at most $\kappa < 0.17$ (proof of Theorem 4.2). So for both activations and all $t > 0$,

$$\big|\sigma(tz + b) - t\,\mathrm{ReLU}(z)\big| \leq |b| + \kappa,$$

and $\sigma(tz + b)/t \to \mathrm{ReLU}(z)$ as $t \to \infty$. Put tx into (2), divide by t, and let $t \to \infty$: the ReLU blocks without biases that have the reads and writes of B_θ and $B_{\theta'}$ satisfy (2). For generic θ these reads and writes are generic, so $B_{\theta'}$ has at least d_{model} more units by Proposition A.11(ii), and for any $M \neq 0$ it is not of the same width, by Theorem 3.1(i). □

Proof of Theorem C.2(iii). The argument also gives Theorem C.1(iii) for ReLU^2 blocks. The construction above gives $3d_{\mathrm{model}}$ units. For the converse, let $B_{\theta'}$, with n more units than B_θ, satisfy (2), and let $r(z) = \max(z, 0)^2$.

Squares. Group the units of both blocks with a nonzero read by the hyperplane $\mathcal{H} = \{x : x\omega + \omega_0 = 0\}$ on which their argument vanishes, with a representative (ω, ω_0) and a side: a unit with read $\lambda\omega$ and bias $\lambda\omega_0$ computes $\lambda^2\, r\big(\pm(x\omega + \omega_0)\big)$ times its write, on side $\pm$ as $\lambda \gtrless 0$. Let $\mu^\pm$ be the sum of λ^2 times the write over the units of $B_{\theta'}$ on side $\pm$, minus the same sum over the units of B_θ. By (2), $B_{\theta'} - B_\theta$ is a polynomial. Near a point of $\mathcal{H}$ on no other hyperplane, the terms of $\mathcal{H}$ are $\mu^+ r(z) + \mu^- r(-z)$, with $z = x\omega + \omega_0$, whose second derivative in the direction of ω jumps across $\mathcal{H}$ by a nonzero multiple of $\mu^+ - \mu^-$. So $\mu^+ = \mu^- =: \mu_{\mathcal{H}}$, the terms of $\mathcal{H}$ are $\mu_{\mathcal{H}}\, z^2$, and

$$\textstyle\sum_{\mathcal{H}} \mu_{\mathcal{H}}\, (x\omega_{\mathcal{H}} + \omega_{0,\mathcal{H}})^2 = xM + \text{const}, \tag{10}$$

with the representatives $(\omega_{\mathcal{H}}, \omega_{0,\mathcal{H}})$.

Costs. For generic θ the units of B_θ have nonzero reads and writes and lie on distinct hyperplanes. Take unit i itself, (w_i, b_i), as the representative of its hyperplane $\mathcal{H}_i$, on side $+$. Then the sums of λ^2 times the writes of the units of $B_{\theta'}$ on $\mathcal{H}_i$ are $\hat{w}_i + \mu_{\mathcal{H}_i}$ on side $+$ and $\mu_{\mathcal{H}_i}$ on side $-$, so $B_{\theta'}$ has at least one unit on $\mathcal{H}_i$, and at least two unless $\mu_{\mathcal{H}_i} \in \{0, -\hat{w}_i\}$. On a hyperplane of no unit of B_θ, it has at least two units if $\mu_{\mathcal{H}} \neq 0$. Let S and U be the sets of units i of B_θ with $\mu_{\mathcal{H}_i} = -\hat{w}_i$ and with $\mu_{\mathcal{H}_i} \notin \{0, -\hat{w}_i\}$, and $\mathcal{N}$ the set of the other hyperplanes with $\mu_{\mathcal{H}} \neq 0$. Then $n \geq |U| + 2|\mathcal{N}|$.

Dimensions. Modulo constants, a polynomial of degree at most two is a pair of a quadratic form and a linear part, in a space of dimension $d_{\mathrm{model}}(d_{\mathrm{model}} + 3)/2$, in which $(x\omega + \omega_0)^2$ becomes $\varphi(\omega, \omega_0) = (\omega\omega^\top, 2\omega_0\omega)$. For each output coordinate o, with $M_{:,o}$ the column o of M, (10) reads

$$\textstyle\sum_{i \in U} \mu_{\mathcal{H}_i, o}\, \varphi(w_i, b_i) + \sum_{\mathcal{H} \in \mathcal{N}} \mu_{\mathcal{H}, o}\, \varphi(\omega_{\mathcal{H}}, \omega_{0,\mathcal{H}}) = \tau_o,$$

where $\tau_o = (0, M_{:,o}) + \sum_{i \in S} \hat{w}_{i,o}\, \varphi(w_i, b_i)$. So the vectors $\tau_1, \ldots, \tau_{d_{\mathrm{model}}}$ and $\varphi(w_i, b_i)$, $i \in U$, lie in the span of $|U| + |\mathcal{N}|$ vectors.

Genericity. For each of the finitely many pairs of disjoint sets S and U of units of B_θ, these $d_{\mathrm{model}} + |U|$ vectors span a space of dimension at least $\min\big(d_{\mathrm{model}}(d_{\mathrm{model}} + 3)/2, d_{\mathrm{model}} + |U|\big)$ for generic θ. The condition is that some minor of the matrix with these columns, a polynomial in θ, does not vanish. It holds when the writes $\hat{w}_i$, $i \in S$, vanish and the reads are generic: then the $\tau_o = (0, M_{:,o})$ span all linear parts, as M is invertible, and for generic columns w_i, any $\min\big(|U|, d_{\mathrm{model}}(d_{\mathrm{model}} + 1)/2\big)$ of the forms $w_i w_i^\top$ are linearly independent.

Conclusion. So $|U| + |\mathcal{N}| \geq \min\big(d_{\mathrm{model}}(d_{\mathrm{model}} + 3)/2, d_{\mathrm{model}} + |U|\big)$. If the minimum is $d_{\mathrm{model}} + |U|$, then $|\mathcal{N}| \geq d_{\mathrm{model}}$ and $n \geq 2d_{\mathrm{model}}$. Otherwise $|U| + |\mathcal{N}| \geq d_{\mathrm{model}}(d_{\mathrm{model}} + 3)/2 \geq 2d_{\mathrm{model}}$, and again $n \geq 2d_{\mathrm{model}}$. For $M \neq 0$ of any rank, the same argument with $\mathrm{rank}\, M$ in place of d_{model} gives $n \geq 2$. □